Weil Conjectures

Chapter 12: The trace formula

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12 The trace formula

Let X_0 be a separated scheme of finite type over $\mathbf{F}_q$, let $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let $\ell \nmid q$. By Lemma 12.5, the points of X_0 over $\mathbf{F}_{q^m}$ are the fixed points of the m-th power of the Frobenius Φ_X on X. The Lefschetz–Verdier formula (Theorem 11.13*) writes a trace on cohomology as a sum of local terms, but it says nothing about what a local term is. To count points we need the local term of Frobenius at a fixed point to be the trace on the stalk.

For a self-map φ of a compact manifold with transverse fixed points, the Lefschetz formula counts each fixed point x with the sign of $\det(1 - d\varphi_x)$. The differential of Frobenius is zero, so we expect every sign to be $+1$. On a curve this can be seen directly. If y is a fixed point of Φ_X^m, u is a uniformizer at y defined over $\mathbf{F}_{q^m}$, and v is the same function on the second factor, then near (y, y) the graph of Φ_X^m is $v = u^{q^m}$, and it meets the diagonal $v = u$ with multiplicity one, since $u^{q^m} - u$ has a simple zero at $u = 0$.

Question 12.1 (The Lefschetz formula for Frobenius)**.** Does the Lefschetz formula hold for Frobenius?

For constant coefficients the local term at a transverse fixed point is 1, so on a proper X transversality would be enough. Two things stand in the way in general. First, X need not be

proper. If we compactify, the new fixed points at the boundary have zero stalk for $j_!K$, and we need their local terms to vanish. Second, a local term is computed from $R\mathcal{H}om(c_2^*K, c_1^!K)$ near the fixed point, where c_1, c_2 are the two projections of the graph of Frobenius to X (Definition 11.5*). The stalk $K_{\bar{y}}$ does not determine this complex (Remark 12.4), so the vanishing of the local term at a point where the stalk vanishes is a real statement, which transversality alone does not give.

One computation on a curve settles both problems: Proposition 12.16 says that for any morphism $\psi : (\Phi_X^m)^*K \to K$ on a smooth curve, the local term at a rational point y is $\operatorname{tr}(\psi_{\bar{y}} \mid K_{\bar{y}})$. The proof, which is Grothendieck's [SGA5, XV, §3], uses $d\Phi_X = 0$ a second time: the graph $v = u^{q^m}$ is tangent to the axis $v = 0$ and the diagonal is not, so blowing up (y, y) separates them, and near the exceptional divisor the local term becomes a restriction map that vanishes when $K_{\bar{y}} = 0$ (Remark 12.6). Since both sides of the formula are additive along an open subscheme and its complement (Lemma 12.19) and compatible with $Rf_!$ (Lemma 12.18), we never need to compactify anything but $\mathbf{A}^1 \subset \mathbf{P}^1$.

The answer is Theorem 12.24: for every Weil complex $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ (Section 10.5*), where $\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$ and A is a finite commutative ℓ-primary ring, and for every $m \geq 1$, we have in Λ

$$\operatorname{tr}_\Lambda(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \operatorname{tr}_\Lambda(\mathrm{Frob}_{y,m} \mid K_{\bar{y}}).$$

Here $\operatorname{tr}_\Lambda$ is the trace of an endomorphism of a perfect complex (Definition 11.1*), and $\mathrm{Frob}_{y,m}$ is the m-th iterate of the Weil structure on the stalk at the fixed point $\bar{y}$. For $K = \Lambda_X$ and Λ a field the formula is Weil's axiom (b), and it gives the rationality of the zeta function (Corollary 12.31).

Purity, however, needs the formula for general K. For $\Lambda \in \{\mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$, in the form of an Euler product (Theorem 12.30), it reads

$$\prod_{x \in |X_0|} \det(1 - T^{\deg x} \mathrm{Frob}_x \mid K_{\bar{x}})^{-1} = \prod_i \det(1 - T \mathrm{Frob}_{X,K} \mid H_c^i(X, K))^{(-1)^{i+1}},$$

which relates the local eigenvalues of K to the eigenvalues on cohomology.

The conventions of Chapter 11* remain in force.

12.1 Frobenius correspondences

Two Frobenius maps act on $X = X_0 \times_{k_0} k$: the k-linear scheme Frobenius Φ_X and the semilinear geometric Frobenius Fr_X that defines Weil structures. We prove the trace formula for Φ_X, whose graph is a correspondence over k, and Lemma 12.3 translates it to Fr_X.

The operator T_{c_m} on cohomology with compact support defined below makes sense even when X is not proper, because the second leg Φ_X^m of the graph c_m is proper. Indeed, Φ_X and all its iterates are finite, since a finitely generated algebra over a perfect field is finite over its subring of q-th powers.

We cannot work with Fr_X directly. Indeed, the Weil structure ϕ_K is a map $\mathrm{Fr}_X^* K \to K$, but Fr_X lies over the automorphism $\mathrm{Spec}(\sigma)$ of s. Hence its graph is not a morphism $X \to X \times_s X$ of k-schemes. Since the formalism of Chapter 11* uses $c_1^!$ only for k-morphisms, it does not apply to this graph. The k-linear $\Phi_X = F_X \circ \mathrm{Fr}_X$, on the other hand, has a graph of the required kind. Pullback along the absolute Frobenius F_X is the identity on étale sheaves, and this is how

Lemma 12.3 turns ϕ_K into a map $\Phi_X^* K \to K$. For example, on $X = \mathbf{A}^1_k$ the map Φ_X is $t \mapsto t^q$, and the fixed points of Φ_X^m are the q^m elements of $\mathbf{F}_{q^m}$. Lemma 12.5 describes them in general.

We allow maps $\Phi_X^* K \to K$ that are not isomorphisms, because the proof over E clears denominators: it multiplies a Weil structure by a power of π_E (Lemma 12.22). The operators below are the two sides of the trace formula in Theorem 12.20.

Definition 12.2 (Frobenius correspondences and their operators)**.** Let X_0/k_0 be separated and of finite type, and put $X = X_0 \times_{k_0} k$. Denote by $\Phi_X : X \to X$ the base change of the q-power scheme Frobenius of X_0. It is a finite k-morphism and a universal homeomorphism. A *Frobenius correspondence* on K is a morphism, not necessarily invertible,

$$\psi : \Phi_X^* K \longrightarrow K, \qquad K \in D^b_{\mathrm{ctf}}(X, A).$$

Its iterates are the morphisms

$$\psi_m = \psi \circ \Phi_X^* \psi \circ \cdots \circ (\Phi_X^{m-1})^* \psi : (\Phi_X^m)^* K \longrightarrow K.$$

Let $c_m = (\mathrm{id}_X, \Phi_X^m) : X \to X \times_s X$ be the graph of Φ_X^m. Since its first leg is the identity, ψ_m is a cohomological correspondence on (X, K) for c_m. Its operator on cohomology with compact support (Definition 11.11*) is the endomorphism

$$T_{c_m}(\psi_m) = R\Gamma_c(X, \psi_m) \circ (\Phi_X^m)^*,$$

where $(\Phi_X^m)^* : R\Gamma_c(X, K) \to R\Gamma_c(X, (\Phi_X^m)^* K)$ is the pullback map induced by the unit of $(\Phi_X^m)^* \dashv R(\Phi_X^m)_*$. For $x \in X_0(\mathbf{F}_{q^m})$ let $\bar{x} : s \to X$ be the associated fixed point. The stalk of ψ_m at $\bar{x}$ is an endomorphism $\psi_{m,\bar{x}} : K_{\bar{x}} \to K_{\bar{x}}$.

The operator $T_{c_m}(\psi_m)$ is defined with the k-linear Φ_X, while the Frobenius operator of Section 10.5*, whose trace the theorem is about, is defined with Fr_X. The next lemma says that the two agree, so that the trace formula proved for Φ_X is a statement about $\mathrm{Frob}_{X,K}$.

Lemma 12.3 (Scheme Frobenius and geometric Frobenius)**.** *Let Fr_X be the semilinear geometric Frobenius of Definition 10.18*, and for a morphism $\phi : \mathrm{Fr}_X^* K \to K$ put $\phi_m = \phi \circ \mathrm{Fr}_X^* \phi \circ \cdots \circ (\mathrm{Fr}_X^{m-1})^* \phi$. There are natural isomorphisms*

$$\kappa_m : (\Phi_X^m)^* K \xrightarrow{\sim} (\mathrm{Fr}_X^m)^* K$$

compatible with iteration. For a Weil structure $\phi : \mathrm{Fr}_X^* K \xrightarrow{\sim} K$, put $\psi = \phi\kappa_1$. Then

$$\psi_m = \phi_m \kappa_m, \qquad T_{c_m}(\psi_m) = \mathrm{Frob}_{X,K}^m,$$

where $\mathrm{Frob}_{X,K}$ is the geometric Frobenius operator on compact cohomology of Definition 10.19. The same identity of operators holds for an arbitrary morphism $(\mathrm{Fr}_X^m)^* K \to K$ in place of ϕ_m.*

Proof. The maps Φ_X and Fr_X differ by the absolute Frobenius, and pullback along the absolute Frobenius is naturally the identity on étale sheaves. This gives κ_m, and we then compare the two pullback maps on one compactification. On coordinate rings the two maps are

$$\Phi_X^\#(a \otimes b) = a^q \otimes b, \qquad \mathrm{Fr}_X^\#(a \otimes b) = a \otimes b^{1/q}$$

so that $\Phi_X = F_X \operatorname{Fr}_X$, where F_X is the absolute q-power Frobenius of X. The absolute Frobenius commutes with every morphism of schemes over $\mathbf{F}_p$, so also $\Phi_X^m = \operatorname{Fr}_X^m F_X^m$. On the classical constructible category, pullback along absolute Frobenius is naturally isomorphic to the identity, because for an étale object its relative Frobenius is an isomorphism, functorial in that object. Transporting this natural isomorphism to $D^b_{\mathrm{ctf}}(X, A)$ by the comparison with the pro-étale site, we obtain $\theta : F_X^* \simeq \mathrm{id}$. It is compatible with pullback along every morphism h, that is, $h^*\theta = \theta h^*$ under $h^*F^* = F^*h^*$, since relative Frobenius of étale objects is compatible with base change. Let $\theta^m : (F_X^m)^* \simeq \mathrm{id}$ be its m-fold composite and put

$$\kappa_m = \theta^m_{(\operatorname{Fr}_X^m)^* K} : (\Phi_X^m)^* K = (F_X^m)^* (\operatorname{Fr}_X^m)^* K \xrightarrow{\sim} (\operatorname{Fr}_X^m)^* K.$$

Compatibility with iteration means that $\kappa_{m+1} = \kappa_{m,\operatorname{Fr}_X^* K} \circ (\Phi_X^m)^* \kappa_1$, and it follows from the compatibility of θ with pullback along Φ_X^m. Now $\psi_m = \phi_m \kappa_m$ follows by induction on m. If it holds for m, then

$$\begin{aligned} \psi_{m+1} &= \psi_m \circ (\Phi_X^m)^* \psi = \phi_m \circ \kappa_m \circ (\Phi_X^m)^* \phi \circ (\Phi_X^m)^* \kappa_1 \\ &= \phi_m \circ (\operatorname{Fr}_X^m)^* \phi \circ \kappa_{m,\operatorname{Fr}_X^* K} \circ (\Phi_X^m)^* \kappa_1 = \phi_{m+1} \kappa_{m+1}, \end{aligned}$$

where the third equality is the naturality of κ_m in K.

For the pullback maps on compact cohomology, let $j : X \hookrightarrow \bar{X}$ be the base change of a compactification of X_0 over k_0, and use the pullback maps pb_h of Section 10.5*. The endomorphisms $\Phi_{\bar{X}}$, $\operatorname{Fr}_{\bar{X}}$ and $F_{\bar{X}}$ of the scheme $\bar{X}$ restrict to Φ_X, Fr_X and F_X. Each of them has inverse image X of X, so each commutes with $j_!$, and $\Phi_{\bar{X}}^m = \operatorname{Fr}_{\bar{X}}^m F_{\bar{X}}^m$. Put $L = j_! K$, so that $R\Gamma_c(X, K) = R\Gamma(\bar{X}, L)$.

First, the map $(\Phi_X^m)^*$ of Definition 12.2 is $\mathrm{pb}_{\Phi_{\bar{X}}^m, L}$. Indeed, Φ_X^m is finite, so $R(\Phi_X^m)_! = R(\Phi_X^m)_*$, and the square $j\Phi_X^m = \Phi_{\bar{X}}^m j$ is cartesian, so $j_! R(\Phi_X^m)_* \simeq R(\Phi_{\bar{X}}^m)_* j_!$. Under this isomorphism the identification $R\Gamma_c(X, R(\Phi_X^m)_* -) = R\Gamma_c(X, -)$ used there is the composition of direct images $R\Gamma(\bar{X}, R(\Phi_{\bar{X}}^m)_* -) = R\Gamma(\bar{X}, -)$, and the unit for Φ_X^m corresponds to the unit for $\Phi_{\bar{X}}^m$.

Second, by transitivity, $\mathrm{pb}_{\Phi_{\bar{X}}^m, L} = \mathrm{pb}_{F_{\bar{X}}^m, (\operatorname{Fr}_{\bar{X}}^m)^* L} \circ \mathrm{pb}_{\operatorname{Fr}_{\bar{X}}^m, L}$. The absolute Frobenius $F_{\bar{X}}^m$ satisfies the hypothesis of [Stacks, 03SP], since its relative Frobenius over each étale object is an isomorphism, functorial in that object. Since A is finite, we may compute on the étale site (Proposition 5.15*). There the formal argument of [Stacks, 03SP] identifies $(F_{\bar{X}}^m)_*$ and $(F_{\bar{X}}^m)^{-1}$ with the identity functors of the topos, compatibly with the unit, and this identification is θ^m, since both come from the relative Frobenius isomorphisms of étale objects. Hence for every $N \in D^b_{\mathrm{ctf}}(\bar{X}, A)$ the pullback map $\mathrm{pb}_{F_{\bar{X}}^m, N}$ followed by $R\Gamma(\bar{X}, \theta^m_N)$ is the identity of $R\Gamma(\bar{X}, N)$. With $N = (\operatorname{Fr}_{\bar{X}}^m)^* L$ this gives

$$R\Gamma_c(X, \kappa_m) \circ (\Phi_X^m)^* = \mathrm{pb}_{\operatorname{Fr}_X^m}.$$

Finally, the naturality of $\mathrm{pb}_{\operatorname{Fr}_X}$ moves the maps $R\Gamma_c(X, (\operatorname{Fr}_X^i)^* \phi)$ past the pullback maps, and transitivity gives $\mathrm{Frob}_{X,K}^m = R\Gamma_c(X, \phi_m) \circ \mathrm{pb}_{\operatorname{Fr}_X^m}$. Hence

$$T_{c_m}(\psi_m) = R\Gamma_c(X, \phi_m) \circ R\Gamma_c(X, \kappa_m) \circ (\Phi_X^m)^* = R\Gamma_c(X, \phi_m) \circ \mathrm{pb}_{\operatorname{Fr}_X^m} = \mathrm{Frob}_{X,K}^m.$$

The same computation with an arbitrary morphism $\chi : (\operatorname{Fr}_X^m)^* K \to K$ in place of ϕ_m gives $T_{c_m}(\chi \kappa_m) = R\Gamma_c(X, \chi) \circ \mathrm{pb}_{\operatorname{Fr}_X^m}$. ∎

For example, if X_0 is smooth and geometrically connected of dimension d and $K = A_X$ with its constant Weil structure, then $T_{c_1}(\psi)$ acts on $H_c^{2d}(X, A) = A(-d)$ by q^d, which agrees with the action of geometric Frobenius on $A(-d)$ recorded after Definition 10.19*.

12.2 Local terms on a curve

The local terms of the trace formula live on the fixed point scheme $\mathrm{Fix}(\Phi_X^m) = \mathrm{Fix}(c_m)$ of the graph c_m. More precisely, a local term is defined on each open and closed part of this scheme that is proper over k. The next lemma shows that the fixed scheme is a finite reduced set of points, one for each $\mathbf{F}_{q^m}$-rational point of X_0, so every such point has its own local term. For Frobenius on a smooth curve we then show that this local term is the trace on the stalk.

On $\mathbf{A}^1$ the fixed scheme of Φ^m is $\operatorname{Spec} k[t]/(t^{q^m} - t)$. Its q^m points are reduced because $t^{q^m} - t$ has derivative -1. This is the algebraic form of the transversality seen at the start of Chapter 12. Reducedness alone does not determine the local term, because the stalk does not determine $R\mathcal{H}om(c_2^*K, c_1^!K)$ near the fixed point (Remark 12.4 below).

Remark 12.4 (Local terms are not computed stalk by stalk)**.** The vanishing of the local term at a rational point y with $K_{\bar{y}} = 0$, proved below, is a geometric fact about the restriction from cohomology with supports in the graph of Φ_X^m to cohomology with supports in y, and it does not follow from a locality property of internal Hom. For example, if $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ and $K = j_!A$, then the stalk of K at the origin is zero, but $R\mathcal{H}om_A(j_!A, j_!A) \simeq Rj_*A$ has stalk A in degree zero there. Hence a local term cannot be computed by commuting a geometric stalk functor past this internal Hom.

Lemma 12.5 (The Frobenius fixed scheme)**.** *For $m \geq 1$, the fixed point scheme of Φ_X^m satisfies*

$$\mathrm{Fix}(\Phi_X^m) \simeq \coprod_{x \in X_0(\mathbf{F}_{q^m})} s.$$

In particular, the fixed scheme is finite, reduced and proper over k.

Proof. We reduce to a closed subscheme of affine space, where the equations $t^{q^m} = t$ have simple roots.

For every open $U_0 \subset X_0$ the restriction to $U = U_0 \times_{k_0} k$ of the fixed scheme of Φ_X^m is the fixed scheme of Φ_U^m, because Φ_X is the base change of a morphism of X_0 and so maps U into itself. As X_0 is quasi-compact, we may therefore assume that X_0 is a closed subscheme of $\mathbf{A}^r_{\mathbf{F}_q}$, with coordinates $t_1, \ldots, t_r$.

In this case the fixed scheme is the pullback of the diagonal along the graph. Since the equations of X_0 have coefficients in $\mathbf{F}_q$, we have $(\Phi_X^m)^\# t_i = t_i^{q^m}$, and the fixed scheme is cut out by the defining equations of X_0 together with $t_i^{q^m} - t_i = 0$. Each polynomial $t^{q^m} - t$ is separable, so the algebra $k[t_1, \ldots, t_r]/(t_i^{q^m} - t_i)_i$ is a product of copies of k indexed by $\mathbf{F}^r_{q^m}$. Its quotient by the defining equations of X_0 retains exactly the factors at the rational solutions of those equations, as asserted. In particular the fixed scheme is a closed subscheme of a finite étale k-scheme, hence étale. The argument uses only the equations of X_0, so X_0 need not be reduced. ∎

Thus every fixed point y is an open and closed point of the fixed scheme, and its local term LT_y is defined.

Remark 12.6 (Outline of the local computation)**.** Let X_0 be a smooth curve and $y \in X_0(\mathbf{F}_{q^m})$. The trace formula on $\mathbf{P}^1$ needs Proposition 12.16 below, which says that the local term at y of any morphism $\psi : (\Phi_X^m)^* K \to K$ is the naive trace $\mathrm{tr}_A(\psi_{\bar{y}} \mid K_{\bar{y}})$, also when K is not lisse at y. To prove it, we filter K by the open complement of y and the point y. By additivity of traces this reduces the proposition to two cases: the point itself, where Lemma 11.10* applies, and the case $K_{\bar{y}} = 0$, where we must show that the local term vanishes.

The second case needs a geometric argument. Indeed, the local term is defined on a neighbourhood of (y, y) in $X \times X$, where the Frobenius graph meets the diagonal, and the vanishing of the stalk at y does not by itself control it. For this we follow [SGA5, III B, §§2–4]. In this approach the local term is a restriction map from the graph to the diagonal, and blowing up (y, y) separates the two curves. After the blowup the local term becomes a restriction to punctured curves near the point where two transverse divisors meet. There it vanishes when the monodromy is an ℓ-group. In general we reduce to that case by passing to finite covers of degree prime to ℓ (Appendix D*); for such a cover the trace map splits the pullback.

What vanishes is a morphism in $D(A)$, before any trace is taken. For this reason the vanishing holds for every ψ, for every finite ℓ-primary ring A, and for every K with $K_{\bar{y}} = 0$. In particular it holds for $j_!K$ on $\mathbf{P}^1$, which is not lisse at infinity. We forget the A-action only in Lemma 12.9, to detect a zero object.

12.2.1 The local term as a restriction map

The vanishing of a local term when $K_{\bar{y}} = 0$ is hard to see from the definition through the trace morphism (Definition 11.8*), so we first replace that definition by a description familiar from topology. There the intersection number of two submanifolds Γ and Δ is computed by restricting the Thom class of Γ to Δ. Since the Thom class is supported on Γ, the result is a class supported on $\Gamma \cap \Delta$. Similarly, in our setting ψ defines a class supported on the graph. A restriction map ρ takes it to cohomology of the diagonal supported at y, and the local term is the evaluation of the result. Note that the map ρ does not involve ψ. It therefore suffices to show that ρ itself is zero when $K_{\bar{y}} = 0$ (Lemma 12.15), and then the vanishing holds for every ψ, invertible or not.

Lemma 12.7 (The local term as a restriction map)**.** *Let X_0 be a smooth curve over k_0, let $X = X_0 \times_{k_0} k$, let $K \in D^b_{\mathrm{ctf}}(X, A)$, and let y be a fixed point of Φ^m_X. Let $\Gamma \subset X \times X$ be the transposed graph of Φ^m_X, the image of the closed immersion $\gamma = (\Phi^m_X, \mathrm{id}_X)$, and let $\Delta \subset X \times X$ be the diagonal, the image of $\delta = (\mathrm{id}_X, \mathrm{id}_X)$. On the strict localization T of $X \times X$ at (y, y) put*

$$P = (\mathbb{D}_X K \boxtimes K)|_T.$$

A morphism $\psi : (\Phi^m_X)^* K \to K$ determines a class $[\psi] \in H^0_\Gamma(T, P)$, and the local term $\mathrm{LT}_y(\psi)$ of Definition 11.9 is the image of this class under the composite*

$$R\Gamma_\Gamma(T, P) \xrightarrow{\rho} R\Gamma_y(\Delta, \delta^* P) \xrightarrow{\mathrm{ev}_K} R\Gamma_y(X, \omega_X) \simeq R\Gamma(y, \omega_y) \xrightarrow{\int_y} A,$$

where ρ is restriction from Γ-supported to y-supported cohomology.

In this lemma the two factors of the product are interchanged compared with the convention $c_m = (\mathrm{id}_X, \Phi^m_X)$ of Definition 12.2.

Proof of Lemma 12.7. We show that the local term and the displayed composite are both adjoint to one map $R\gamma_*\gamma^! P \to R\delta_*\omega_X$. Once the class $[\psi]$ and the map ρ are built, the comparison is formal, and computing on étale models changes nothing.

By external duality, $\gamma^! P$ is identified with $R\mathcal{H}om((\Phi^m_X)^* K, K)$, so exceptional adjunction identifies ψ with $[\psi]$. The map ρ is constructed from the two canonical arrows

$$R\gamma_*\gamma^! P \longrightarrow P \longrightarrow R\delta_*\delta^* P.$$

After restriction to Δ, the source is supported on $\Gamma \cap \Delta$. By Lemma 12.5 the scheme $\Gamma \cap \Delta$ is reduced, and on T it is the single point (y, y). Hence the resulting map factors through $R\Gamma_y(\Delta, \delta^* P)$. Equivalently, it is the ordinary pullback of the object with supports, followed by the base change map to local cohomology with support in the intersection.

The adjoint of $\mathrm{ev}_K : \delta^* P \to \omega_X$ is the map $P \to R\delta_* \omega_X$ used to define the trace morphism in Definition 11.8*. Hence the displayed composite, before $\int_y$, is adjoint to the map

$$R\gamma_* \gamma^! P \longrightarrow P \longrightarrow R\delta_* \omega_X .$$

The local term is built from the same map. Indeed, by exceptional proper base change for δ, the adjoint of this map, which is supported on y, is the trace map $\gamma^! P \to Ri_* \omega_y$ of Definition 11.9*. The triangle identities of the ordinary and exceptional adjunctions then show that the two constructions give the same element of A.

The constructions above also make sense on the constructible étale models, where restriction to the strict localization is the filtered limit over étale neighbourhoods. The comparison theorems identify inverse images, exceptional adjunctions, duality and the support maps on the two sites, and hence they identify the class obtained on the étale models with the pro-étale local term of Definition 11.9*. ∎

12.2.2 Vanishing on a strictly local surface

The vanishing of the local term is explained by the position of the curves at (y, y). Since $d\Phi_X = 0$, the transposed graph Γ is tangent to the vertical axis $V = \{y\} \times X$. The diagonal Δ and the horizontal axis H have other tangent directions. This persists on suitable covers of degree prime to ℓ (Lemma 12.13 below). Blowing up the closed point separates the curves: the strict transforms of $\Gamma \cup V$, of Δ and of H meet the exceptional divisor $E \simeq \mathbf{P}^1$ in different points.

Let M be the direct image of the coefficients from the complement of E. We consider M on E minus the points of $\Gamma \cup V$, and let z be the point where H meets E. There M is locally constant away from z, and its stalk at z is zero. Both facts use that the monodromy is an ℓ-group (Lemma 12.9). Now a branch of Δ meets E in one point w, and the restriction to this branch factors through the stalk M_w. Composing the restriction from E to M_w with the specialization isomorphism from M_w to the generic stalk of E, we get the restriction to the zero stalk M_z followed by specialization. Hence the restriction to M_w is zero.

The transverse case is Lemma 12.9, and Lemma 12.11 reduces to it by blowing up the closed point. We compute on classical étale models of strictly local schemes, which are filtered limits of schemes of finite type, and in the proof of Lemma 12.15 we transport the result to the pro-étale site.

Lemma 12.8 (A punctured surface and a punctured curve)**.** *Let S be the strict localization of a smooth surface over an algebraically closed field k of characteristic p at a closed point z, let $\ell \neq p$, and let $D_1, D_2 \subset S$ be regular divisors meeting transversely at z. Then the restriction map*

$$R\Gamma(S - D_2, \mathbf{F}_\ell) \longrightarrow R\Gamma(D_1 - z, \mathbf{F}_\ell)$$

is an isomorphism.

Proof. Both complexes have cohomology $\mathbf{F}_\ell$ in degree zero, $\mathbf{F}_\ell(-1)$ in degree one, and zero in all other degrees. For the surface this follows from localization, the acyclicity of the strictly local

schemes S, D_2, and the purity identity $i^!_{D_2}\mathbf{F}_\ell = \mathbf{F}_\ell(-1)[-2]$. This identity, in turn, comes by transitivity from the dualizing complexes of the smooth schemes S and D_2.

For the punctured trait $D_1 - z$, vanishing in degrees ≥ 2 follows from Proposition D.5*, by passing to the filtered limit of punctured neighbourhoods that are curves. In degree one the Kummer sequence gives $H^1(-, \mu_\ell) = k'^*/k'^{*\ell}$, where k' is the fraction field. Moreover, the units of the strictly henselian trait are ℓ-divisible. Hence the valuation induces $k'^*/k'^{*\ell} \simeq \mathbf{Z}/\ell$, and the class of a uniformizer is a generator.

To see that the restriction map itself, and not only the two sides, is an isomorphism, we find a generator of H^1 on each side. The restriction is clearly the identity in degree zero. In degree one, let g be a local equation of D_2. Since the regular local ring of S is factorial, $\mathrm{Pic}(S - D_2) = 0$ and the units on $S - D_2$ are the units of S times the powers of g. As the units of S are ℓ-divisible, the Kummer sequence shows that $H^1(S - D_2, \mu_\ell) \simeq \mathbf{Z}/\ell$, generated by the class of g. Since D_1 and D_2 meet transversely, g restricts to a uniformizer of D_1, whose class generates $H^1(D_1 - z, \mu_\ell)$. So the restriction is an isomorphism in degree one as well, and since both complexes vanish in the other degrees, it is an isomorphism. ∎

Lemma 12.9 (Cohomology near two transverse divisors)**.** *Let X be a smooth surface over an algebraically closed field of characteristic p, or an étale limit with affine transition maps of such surfaces. Let D_1, D_2 be regular divisors meeting transversely at exactly one point z, and let*

$$j : X - D_2 \hookrightarrow X, \qquad j' : X - (D_1 \cup D_2) \hookrightarrow X - D_2.$$

Let $\ell \neq p$, let A be a finite commutative ℓ-primary ring, and let N be a bounded complex of A-modules on $X - (D_1 \cup D_2)$ with finite locally constant cohomology whose monodromy factors through a finite ℓ-group. Put $M = Rj_*j'_!N$. Then $M_z = 0$, and $M|_{D_2 - z}$ has finite locally constant cohomology. If D_2 is connected, then for every closed geometric point $w \in D_2 - z$ the restriction map

$$R\Gamma(D_2, M|_{D_2}) \longrightarrow M_w$$

is zero in $D(A)$.

Proof. The idea is that, since the monodromy is an ℓ-group, we may filter the coefficients until they become the constant sheaf $\mathbf{F}_\ell$. For constant coefficients, $M_z = 0$ is a statement on the strict localization of X at z: restriction from the complement of D_2 to the punctured curve $D_1 - z$ is an isomorphism on cohomology. This holds because a local equation of D_2 gives the generator of H^1 on both sides. Once we know that M is locally constant along $D_2 - z$, the vanishing of the restriction map follows from $M_z = 0$ by specialization along D_2.

For the assertion at z, pass to the strict localization S of X at z and write $D_{1,S}, D_{2,S}$ for the two transverse divisors. By the formula for stalks of a direct image along an open immersion [Stacks, 03Q7] and localization we have

$$M_z = R\Gamma(S - D_{2,S}, j'_!N).$$

To prove that this object is zero we may forget the A-action, since the exact forgetful functor is conservative and the cohomology of sheaves of modules agrees with the cohomology of the underlying abelian sheaves [Stacks, 03FD]. A finite abelian ℓ-group with an action of a finite ℓ-group has a finite filtration whose quotients are $\mathbf{F}_\ell$ with trivial action. To see this, filter by

powers of ℓ and use the nilpotence of the augmentation ideal of the group algebra over $\mathbf{F}_\ell$. Since $j'_!$ and $R\Gamma(S - D_{2,S}, -)$ are exact, the class of N for which $R\Gamma(S - D_{2,S}, j'_!N) = 0$ is closed under extensions. Using truncations and these composition series we reduce the assertion to $N = \mathbf{F}_\ell$ on $S - (D_{1,S} \cup D_{2,S})$.

By Lemma 12.8, applied to S with the divisors $D_{1,S}, D_{2,S}$, the restriction map $R\Gamma(S - D_{2,S}, \mathbf{F}_\ell) \to R\Gamma(D_{1,S} - z, \mathbf{F}_\ell)$ is an isomorphism. The localization sequence for $D_{1,S} - z$ and its complement in $S - D_{2,S}$ identifies $R\Gamma(S - D_{2,S}, j'_!\mathbf{F}_\ell)$ with the cone of this map, shifted by -1. Hence this complex is zero, and $M_z = 0$ in $D(A)$ by conservativity.

We turn to the local constancy of M along $D_2 - z$, where only the divisor D_2 remains. Since D_1 meets D_2 only at z, we have $j'_!N = N$ near $D_2 - z$, so $M = Rj_*N$ there. To compute Rj_*N we use Abhyankar's lemma along the regular divisor D_2 [Stacks, 0EYG]: étale locally on D_2, a finite tame local system is trivialized by a Kummer cover of a local parameter of D_2 and an unramified cover. The lemma applies here because our finite ℓ-group monodromy is tame, wild inertia being pro-p. Consequently, the stalks of Rj_*N are the continuous cohomology of the same representation of tame inertia, and their specialization maps along $D_2 - z$ are isomorphisms. Explicitly, the pro-ℓ part of tame inertia is $\mathbf{Z}_\ell(1)$. Its cohomology with coefficients in a finite ℓ-primary module is computed by the two-term complex given by a topological generator minus one. On the other hand, the finite quotients of the remaining factors of tame inertia have order prime to ℓ, so taking invariants under them is exact on these coefficients. Finally, we may choose the parameter and the unramified cover on an étale neighbourhood. Hence this description, including the specialization maps, is locally constant. Applying the argument to the cohomology sheaves of N and using the bounded spectral sequence proves the assertion for N.

The last assertion combines the two facts just proved. A global section of $M|_{D_2}$ specializes to every stalk. Its image at z is zero, and along $D_2 - z$ specialization is an isomorphism on stalks. To make this precise, choose a geometric generic point $\bar{\eta}$ of the connected regular curve D_2, together with specialization maps from $\bar{\eta}$ to z and to w. By naturality of restriction we get a commutative square

$$\begin{array}{ccc} R\Gamma(D_2, M|_{D_2}) & \longrightarrow & M_w \\ \downarrow & & \downarrow\sim \\ M_z = 0 & \longrightarrow & M_{\bar{\eta}}. \end{array}$$

The right arrow is an isomorphism by the local constancy just proved, so the upper arrow is zero. ∎

Since the restriction to a curve meeting D_2 in one point factors through the stalk of M at that point, the lemma implies that restrictions to punctured curves vanish.

Lemma 12.10 (Restriction to a punctured curve)**.** *In the situation of Lemma 12.9 with D_2 connected, let $C \subset X$ be a closed curve with $C \cap D_2 = \{w\}$, and let $C_{(w)}$ be its strict localization at w. Then the restriction map*

$$R\Gamma(X - D_2, j'_!N) \longrightarrow R\Gamma(C_{(w)} - w, (j'_!N)|_{C_{(w)}-w})$$

is zero in $D(A)$.

Proof. The source is $R\Gamma(X - D_2, j'_!N) = R\Gamma(X, M)$, by the definition $M = Rj_*j'_!N$. Since C meets D_2 only at w, the punctured curve $C_{(w)} - w$ lies over the strict localization of X at w and avoids D_2. The map therefore factors as the restriction to $R\Gamma(D_2, M|_{D_2})$, followed by

the restriction to the stalk $M_w = R\Gamma(X_{(w)} - D_2, j_!'N)$ (by the stalk formula), followed by the restriction to the displayed target. Since all these maps are restrictions of the same complex, the composite is the given map. If $w \neq z$, the middle map is zero by Lemma 12.9. If $w = z$, the intermediate object M_z is zero. ∎

Near (y, y) the curves we care about do not meet transversely, since the graph, the diagonal and the axis H all pass through the same point. The next lemma reduces this configuration to the transverse one by a single blowup.

Lemma 12.11 (Vanishing of restrictions after blowing up)**.** *Let T be the strict localization of a smooth surface at a closed point t. Let H be a regular curve through t, and let D and C be reduced closed curves through t such that the tangent cone of D at t meets the tangent cones of C and of H only in 0. Let $j' : T - (D \cup H) \hookrightarrow T - D$ be the open immersion and let N be a bounded complex on $T - (D \cup H)$ with finite locally constant cohomology and finite ℓ-group monodromy. Then the restriction map*

$$R\Gamma(T - D, j_!'N) \longrightarrow R\Gamma(C - t, (j_!'N)|_{C-t})$$

is zero in $D(A)$.

Proof. After blowing up t, the three curves are separated along the exceptional divisor, which then plays the role of D_2 in the previous lemma.

Let $\widetilde{T} \to T$ be the blowup in t, with exceptional divisor $E \simeq \mathbf{P}^1_k$, and let D', H', C' be the strict transforms. In these terms, the hypotheses on tangent cones say that $D' \cap E$ is disjoint from both $C' \cap E$ and $H' \cap E$. We therefore apply Lemmas 12.9 and Lemma 12.10 to the surface $X = \widetilde{T} - D'$ with the divisors $D_1 = H' \cap X$ and $D_2 = E \cap X = E - D'$. Here the divisor $E - D'$ is connected, since it is a projective line with finitely many points removed, and $H' \cap X$ meets it transversely in the single point z corresponding to the tangent direction of H. Since the blowup is an isomorphism off E, the source of the map is $R\Gamma(X - E, j_!'N')$ for the pulled-back complex N'.

The strict transform C' is proper and quasi-finite over the strictly local curve C, hence finite. Its closed fibre is the finite set of tangent directions of its branches, so C' is a finite disjoint union of strictly local curves C'_w, each meeting E at a single point w of $E - D'$. Each restriction to $C'_w - w$ is zero by Lemma 12.10. Since $C' - E \simeq C - t$, the given restriction map is the direct sum of these zero maps. ∎

12.2.3 The local term at a zero stalk

In the setting of Lemma 12.7 we compute on the classical étale models. This is legitimate because the classical comparison identifies tensor products, duality, exceptional adjunctions and the pairing between graph and diagonal with their versions there. Let X_1 and X_2 be two copies of the strict localization of X at y. The strict localization of $X_1 \times X_2$ at (y, y) is the strictly local surface T of Lemma 12.7. Write

$$H = X_1 \times \{y\}, \qquad V = \{y\} \times X_2,$$

and keep Γ and Δ for the transposed Frobenius graph and the diagonal in T. Let u and v be the pullbacks to the two factors X_1 and X_2 of the same uniformizer of X at y, so that Δ, V and H have equations $u - v = 0$, $u = 0$ and $v = 0$. Write $\varphi(v) = (\Phi_X^m)^{\#}u$, a power series in v; the graph

Γ has equation $u - \varphi(v) = 0$. Since $d\Phi_X = 0$, the graph Γ is tangent to V, while Δ and H have other tangent directions.

The next three lemmas prepare the proof of Lemma 12.15.

Lemma 12.12 (Covers of degree prime to ℓ)**.** *Let S be the strict localization of a smooth curve over the algebraically closed field k at a closed point s, and let $\mathcal{F}_1, \dots, \mathcal{F}_r$ be finite locally constant sheaves on $(S - s)_{\text{ét}}$. There is an integer d_1 prime to ℓ with the following property. For every multiple d of d_1 that is prime to ℓ there is a finite flat morphism $S' \to S$ of degree d such that*

1. *S' is the strict localization of a smooth curve over k at a closed point s', and s' is the only point of S' over s; more precisely, $S' \to S$ is the base change of a finite flat morphism $C' \to C$ of smooth curves over k, where C is an étale neighbourhood of s, which is étale away from s and has a single point over s,*
2. *$S' - s' \to S - s$ is finite étale, and*
3. *the monodromy of the pullback of each $\mathcal{F}_i$ to $S' - s'$ factors through a finite ℓ-group.*

Proof. We take the fixed field of an ℓ-Sylow subgroup of the monodromy, adjust the degree by an Eisenstein extension, and descend to curves.

Let F be the fraction field of the discrete valuation ring of S. Then the punctured trait $S - s$ is $\operatorname{Spec}(F)$, so the $\mathcal{F}_i$ are finite sets with a continuous action of $\operatorname{Gal}(F^{\text{sep}}/F)$. We choose a finite Galois extension L/F over which all of them become constant, let P be an ℓ-Sylow subgroup of $G = \operatorname{Gal}(L/F)$, and put $F_1 = L^P$ and $d_1 = [F_1 : F] = [G : P]$, which is prime to ℓ.

Let $d = d_1 e$ be a multiple of d_1 prime to ℓ. The normalization S_1 of S in the separable extension F_1 is finite over S, and it is local because it is a domain finite over a henselian local ring. Hence S_1 is a strictly henselian discrete valuation ring with residue field k, and it has degree d_1 over S. If $e > 1$, let t be a uniformizer of S_1. The Eisenstein polynomial $z^e + tz - t$ is irreducible, and its derivative $ez^{e-1} + t$ is nonzero, also when $p \mid e$, so it is separable. Hence $S' = S_1[z]/(z^e + tz - t)$ is a discrete valuation ring, totally ramified of degree e over S_1, with separable fraction field F' over F. If $e = 1$ we put $S' = S_1$. Then $S' \to S$ is finite flat of degree d, with a single point s' over s, and $S' - s' = \operatorname{Spec}(F') \to \operatorname{Spec}(F)$ is finite étale. The group $\operatorname{Gal}(F^{\text{sep}}/F')$ acts on the stalks of the $\mathcal{F}_i$ through its image in G, which lies in $\operatorname{Gal}(L/F_1) = P$ since $F_1 \subset F'$.

Finally, by a limit argument $S' \to S$ descends to a finite flat morphism $C' \to C$ onto an étale neighbourhood C of s, which we may take with normal source. Since k is perfect, the normal curve C' is smooth. Moreover, the morphism $C' \to C$ is étale over a dense open of C, and after shrinking C around s it is étale away from s. Since k is algebraically closed, $C' \times_C S$ is the product of the strict localizations of C' at the points over s. As it equals the local scheme S', there is a single point over s, and S' is the strict localization of C' there. ∎

Lemma 12.13 (Tangent cones on the cover)**.** *For $i = 1, 2$ let $X_i' \to X_i$ be a finite flat morphism of degree d as in Lemma 12.12, with y_i' the point over y. Let T' be the strict localization of $X_1' \times X_2'$ at $y' = (y_1', y_2')$, and write Γ', V', Δ', H' for the inverse images of Γ, V, Δ, H in T'. Then T' is the strict localization of a smooth surface over k at a closed point, the morphism $T' \to T$ is finite flat of degree d^2 and étale over $T - (H \cup V)$, the reduction H'_{red} is regular, and the tangent cone of $(\Gamma' \cup V')_{\text{red}}$ at y' meets the tangent cones of Δ'_{red} and of H'_{red} only in 0.*

Proof. The two covers descend to finite flat morphisms $C_i' \to C_i$ of degree d between smooth curves, where C_i is an étale neighbourhood of y in X, étale away from y and with a single point y_i' over y (Lemma lemma-covers-degree-prime-to-ell). Their product $C_1' \times C_2' \to C_1 \times C_2$ is a finite flat morphism of degree d^2 of smooth surfaces, étale away from the two fibres through

y, with y' the only point over (y,y). Base change to the strict localization T of $C_1 \times C_2$ at (y,y) gives a scheme finite over T with one point over the closed point, hence local and strictly henselian, and this is T'. This proves the assertions about T' and $T' \to T$.

Choose uniformizers ξ of X_1' and η of X_2', so that the completed local ring of T' at y' is $k[[\xi,\eta]]$. Since the residue field k is algebraically closed, both covers are totally ramified, so u is a unit of $k[[\xi]]$ times ξ^d, and similarly for v. Multiplying ξ and η by suitable dth roots of nonzero constants, we may assume that

$$u \equiv \xi^d, \qquad v \equiv \eta^d \pmod{(\xi,\eta)^{d+1}}.$$

Since Φ_X^m is purely inseparable of degree q^m and fixes y, we have $\operatorname{ord}_v \varphi = q^m > 1$, and hence $\varphi(v) \in (\eta)^{dq^m} \subseteq (\xi,\eta)^{d+1}$. It follows that the pulled-back equations of Γ', V', Δ' and H' have initial forms

$$\xi^d, \qquad \xi^d, \qquad \xi^d - \eta^d, \qquad \eta^d,$$

respectively. The support of the tangent cone of a curve is the zero set of the initial form of its equation, and it does not change when we pass to the reduction, since if J is the radical of I and $J^n \subset I$ then the ideals of initial forms satisfy $\operatorname{in}(J)^n \subset \operatorname{in}(I) \subset \operatorname{in}(J)$. Thus the tangent cone of $(\Gamma' \cup V')_{\mathrm{red}}$ is supported on the line $\xi = 0$, that of H'_{red} on $\eta = 0$, and that of Δ'_{red} on the lines $\xi = \zeta\eta$ with $\zeta^d = 1$. This holds also when $p \mid d$ and $\xi^d - \eta^d$ is a pth power. Away from the origin the first support meets neither of the others. Finally H'_{red} is the curve $\eta = 0$, which is regular. ∎

Lemma 12.14 (Descent of vanishing along a finite étale cover)**.** *Let $f : W' \to W$ be a finite étale morphism of schemes of constant degree n, with n invertible in A, and let $L \in D(W_{\acute{e}t}, A)$. Then the pullback map $R\Gamma(W,L) \to R\Gamma(W', f^{-1}L)$ has a left inverse in $D(A)$. In particular, a morphism $\alpha : M \to R\Gamma(W,L)$ in $D(A)$ is zero if its composite with this pullback map is zero.*

Proof. Since f is finite étale, $f_* = f_!$ is exact, and the trace map $f_* f^{-1}\mathcal{F} \to \mathcal{F}$ composed with the adjunction unit $\mathcal{F} \to f_* f^{-1}\mathcal{F}$ is multiplication by n for every sheaf of A-modules $\mathcal{F}$ [Stacks, 03SH]. Since both maps are natural and the functors are exact, applying them termwise to a complex representing L gives maps $L \to f_* f^{-1} L \to L$ in $D(W_{\acute{e}t}, A)$ whose composite is n. Applying $R\Gamma(W, -)$ and using $R\Gamma(W, f_* f^{-1} L) = R\Gamma(W', f^{-1}L)$, we see that n^{-1} times the map induced by the trace is a left inverse of the pullback. ∎

The map ρ of Lemma 12.7 depends on K and on the geometry of Γ and Δ near (y,y), but not on ψ. We therefore state the vanishing of local terms at a zero stalk as a statement about ρ alone.

Lemma 12.15 (Vanishing of the local term at a zero stalk)**.** *In the setting of Lemma 12.7, let y be a fixed point of Φ_X^m, and suppose $K_{\bar{y}} = 0$. Then*

$$\rho : R\Gamma_\Gamma(T, P) \longrightarrow R\Gamma_y(\Delta, \delta^* P)$$

is zero in $D(A)$. Consequently $\mathrm{LT}_y(\psi) = 0$ for every morphism $\psi : (\Phi_X^m)^* K \to K$.

Proof. We use the notation introduced before Lemma 12.12 and compute on the étale model: we identify ρ with a restriction map r away from y, kill r on a cover with ℓ-group monodromy, and descend.

Step 1: reduction to r. Let $N = (\mathbb{D}_X K)|_{X_1 - y}$, the dual of $K|_{X_1 - y}$ with respect to the restriction of ω_X, and let $Q = K|_{X_2 - y}$. Write $j_1 : X_1 - y \hookrightarrow X_1$ and $j_2 : X_2 - y \hookrightarrow X_2$ for the inclusions. As $K_{\bar{y}} = 0$, localization gives $K|_{X_2} = j_{2!}Q$, and $(\mathbb{D}_X K)|_{X_1} = Rj_{1*}N$ because the dual of an extension by zero is the direct image of the dual. The ordinary external Künneth comparison therefore identifies $P = (\mathbb{D}_X K \boxtimes K)|_T$ with $Rj_* j'_!(N \boxtimes Q)$, where

$$j : T - V \hookrightarrow T, \qquad j' : T - (H \cup V) \hookrightarrow T - V.$$

Both $R\Gamma(T, P)$ and $R\Gamma(\Delta, \delta^* P)$ are zero, because the schemes are strictly local and the stalk at the closed point is $(\mathbb{D}_X K)_{\bar{y}} \otimes^{\mathbf{L}} K_{\bar{y}} = 0$. So the localization triangles identify ρ, shifted by one, with the ordinary restriction $R\Gamma(T - \Gamma, P) \to R\Gamma(\Delta - y, \delta^* P)$, which under the Künneth identification is

$$r : R\Gamma(T - (V \cup \Gamma), j'_!(N \boxtimes Q)) \longrightarrow R\Gamma(\Delta - y, (N \boxtimes Q)|_{\Delta - y}).$$

Step 2: the restriction vanishes on a cover. Choose bounded finite projective lisse models of N and Q by Lemma 3.13*; their terms are finite locally constant, as a punctured strictly local trait is its generic point. Then Lemma 12.12, applied to the terms of the model of N on $X_1 - y$ and to those of Q on $X_2 - y$, gives integers d_1, d_2 prime to ℓ. With $d = \operatorname{lcm}(d_1, d_2)$ we then obtain covers $X'_1 \to X_1$ and $X'_2 \to X_2$ of the same degree d. We want equal degrees so that u and v become dth powers for the same d in Lemma 12.13. Let T', Γ', V', Δ', H' be as in that lemma, and let N' and Q' be the pullbacks of N and Q to $X'_1 - y'_1$ and $X'_2 - y'_2$. By construction, the complex $N' \boxtimes Q'$ has finite ℓ-group monodromy on $T' - (H' \cup V')$, hence also on the smaller open set obtained by removing Γ'. Then the restriction r' on T', defined like r, is zero by Lemma 12.11, applied to T' with $D = (\Gamma' \cup V')_{\mathrm{red}}$, the regular curve H'_{red} and $C = \Delta'_{\mathrm{red}}$; its hypotheses hold by Lemma 12.13, and passing to reductions changes neither the open complements nor the étale cohomology.

Step 3: descent. The map r followed by pullback to $\Delta' - y'$ equals pullback to T' followed by $r' = 0$, because pullback along $T' \to T$ commutes with restriction and with extension by zero. As Δ meets $H \cup V$ only at y, Lemma 12.13 shows that $\Delta' - y' \to \Delta - y$ is finite étale of degree d^2. This degree is invertible in A, so Lemma 12.14 gives $r = 0$, and hence $\rho = 0$. By the comparison theorems, this identity on the classical étale models of these strictly local schemes also holds in the pro-étale formulation. Finally, by Lemma 12.7, $\mathrm{LT}_y(\psi)$ is the image of $[\psi]$ under a composite that begins with ρ, so it is zero. ∎

12.2.4 The local term at a fixed point

The point y is stable under Φ_X^m, so additivity of trace classes applies to the localization triangle for $X \setminus \{y\}$ and $\{y\}$. Here the open piece has zero stalk at y, so its local term vanishes by Lemma 12.15. The closed piece, in turn, is pushed forward from the point, where the local term is a trace by Lemma 11.10*. Note that the argument never uses that K is lisse at y. This is why, in the proof of Theorem 12.20, we can apply the result to $j_! K$ at the point at infinity of $\mathbf{P}^1$.

Proposition 12.16 (The Frobenius local term on a curve)**.** *Let X_0 be a smooth curve over k_0, let $X = X_0 \times_{k_0} k$, let $K \in D^b_{\mathrm{ctf}}(X, A)$ and let $\psi : (\Phi_X^m)^* K \to K$ be any morphism. For $y \in X_0(\mathbf{F}_{q^m})$ one has*

$$\mathrm{LT}_y(\psi) = \mathrm{tr}_A(\psi_{\bar{y}} \mid K_{\bar{y}}).$$

Proof. Let $i : \{y\} \hookrightarrow X$ and $j : U = X \setminus \{y\} \hookrightarrow X$. We split ψ along the localization triangle and compute the two local terms separately.

Splitting ψ. Both pieces are stable under Φ_X^m, and in fact $((\Phi_X^m)^{-1}\{y\})_{\mathrm{red}} = \{y\}$. In the enhanced derived category, the functorial localization triangle is a two-step filtered object,

$$K_U = j_! j^* K \longrightarrow K \longrightarrow i_* i^* K \xrightarrow{+1},$$

and ψ induces a coherent map from the $(\Phi_X^m)^*$-pullback of this filtration to the filtration itself, with induced correspondences ψ_U and $i_*\psi_{\bar{y}}$ on the pieces. This rests on two base change identities. For the open part, $(\Phi_X^m)^* j_! \simeq j_!(\Phi_X^m|_U)^*$ by base change along inverse images in the cartesian square. The square for the point, by contrast, is cartesian only after passing to reductions, so there proper base change followed by nilpotent invariance gives $(\Phi_X^m)^* i_* \simeq i_*(\Phi_X^m|_y)^*$. Both comparisons are mates of the adjunction units for j and i used in the triangle, which is why the induced maps are compatible with the filtration.

Additivity. The second half of Proposition 11.19* applies to the graph c_m, because its hypothesis $\Phi_X^m(\{y\}) \subset \{y\}$ holds for the fixed point y. It gives

$$\mathrm{LT}_y(\psi) = \mathrm{LT}_y(\psi_U) + \mathrm{LT}_y(i_*\psi_{\bar{y}}).$$

The two local terms. The first term is zero by Lemma 12.15, since $(K_U)_{\bar{y}} = 0$. For the second, the description of the closed piece u_Z in Proposition 11.19* shows that $i_*\psi_{\bar{y}}$ is the pushforward along i of the cohomological correspondence $\psi_{\bar{y}}$ on $(y, K_{\bar{y}})$ for the identity correspondence of y. Its local term is therefore $\mathrm{tr}_A(\psi_{\bar{y}} \mid K_{\bar{y}})$, the trace on the perfect complex $K_{\bar{y}}$, by the compatibility with proper pushforward in Theorem 11.13* and Lemma 11.10*. ∎

For $K = A_X$ on $X = \mathbf{A}^1_k$ the proposition gives $\mathrm{LT}_y(\psi_m) = 1$ at each of the q^m fixed points of Φ_X^m. This matches the intersection count at the start of the chapter. For general K the proposition is not a formal consequence of the definitions, as Remark 12.4 explains.

Example 12.17 (Frobenius on the projective line)**.** Let $X = \mathbf{P}^1_k$, $K = A_X$, and let ψ be the constant Frobenius correspondence, so that ψ_m is the identity on every stalk. Both sides of the Lefschetz–Verdier formula for the graph c_m can be computed explicitly. The fixed points of Φ_X^m are the $q^m + 1$ points of $\mathbf{P}^1(\mathbf{F}_{q^m})$. By the proposition each has local term $\mathrm{tr}_A(1 \mid A) = 1$, so the local side is $q^m + 1$. On the global side, $R\Gamma(\mathbf{P}^1, A) = A \oplus A(-1)[-2]$. By Lemma 12.3 the operator $T_{c_m}(\psi_m)$ is the m-th power of geometric Frobenius. It acts by 1 on H^0 and by q^m on $H^2 = A(-1)$, and both degrees are even, so its trace is $1 + q^m$ with no signs. Compare the identity correspondence at the start of Section 11.4*. There the fixed locus is all of $\mathbf{P}^1$ and the answer 2 is a self-intersection. Here the fixed locus is finite and each point counts once.

12.3 Frobenius and $Rf_!$

The trace formula with finite coefficients is proved by induction on the number of affine coordinates. We project a closed subscheme of $\mathbf{A}^d$ to $\mathbf{A}^{d-1}$, apply the induction hypothesis to $Rf_!K$, and identify each stalk of $Rf_!K$ at a rational point with the cohomology of the fibre, a closed subscheme of $\mathbf{A}^1$ over a larger finite field. For this, $Rf_!K$ must carry a Frobenius correspondence with two properties: its operator on $R\Gamma_c$ is that of ψ, and its stalk maps are the Frobenius operators on the cohomology of the fibres.

The square formed by f and the two Frobenii commutes, but it is usually not cartesian. Indeed, already for $f : \mathbf{A}^1 \to s$ the Frobenius of s is the identity, so the fibre product is $\mathbf{A}^1$ itself,

while $\Phi_{\mathbf{A}^1}$ has degree q. The difference is measured by the relative Frobenius. It is a universal homeomorphism, so pullback along it is an equivalence on étale sheaves, and the comparison map built from it is an isomorphism.

Lemma 12.18 (Comparison via relative Frobenius)**.** *Let $f_0 : X_0 \to Y_0$ be a morphism of separated schemes of finite type over k_0, write $f : X \to Y$ for its geometric base change, and let $m \geq 1$. Put*

$$X'_m = Y \times_{\Phi_Y^m, Y, f} X, \quad p_m : X'_m \to X, \quad f'_m : X'_m \to Y, \quad r_m = (f, \Phi_X^m) : X \to X'_m,$$

where p_m and f'_m are the two projections.

1. *The relative Frobenius r_m is a finite universal homeomorphism, $p_m r_m = \Phi_X^m$ and $f'_m r_m = f$, and there are natural isomorphisms*

$$\begin{aligned}\theta_{f,m} : (\Phi_Y^m)^* R f_! K &\simeq R f'_{m!} p_m^* K \xrightarrow{R f'_{m!} \eta_{r_m}} R f'_{m!} R r_{m*} r_m^* p_m^* K \\ &\simeq R f_! (\Phi_X^m)^* K,\end{aligned}$$

   compatible with iteration in m.
2. *For a Frobenius correspondence $\psi : \Phi_X^* K \to K$ on $K \in D^b_{\mathrm{ctf}}(X, A)$, the object $L = R f_! K$ of $D^b_{\mathrm{ctf}}(Y, A)$ carries the Frobenius correspondence $\psi' = R f_! \psi \circ \theta_{f,1} : \Phi_Y^* L \to L$, and its iterates are*

$$\psi'_m = R f_! \psi_m \circ \theta_{f,m}.$$

3. *The composition isomorphism $R\Gamma_c(Y, L) \simeq R\Gamma_c(X, K)$ intertwines the endomorphisms $T_{c_m}(\psi'_m)$ on Y and $T_{c_m}(\psi_m)$ on X of Definition 12.2.*
4. *At $y \in Y_0(\mathbf{F}_{q^m})$, base change for compact supports identifies $(L_{\bar{y}}, \psi'_{m,\bar{y}})$ with*

$$\left(R\Gamma_c(X_{\bar{y}}, K|_{X_{\bar{y}}}), T_{c_1}(\psi_m|_{X_{\bar{y}}})\right),$$

   where the fibre is regarded as a scheme over $\mathbf{F}_{q^m}$, with the q^m-power Frobenius as its scheme Frobenius.

Proof. The only geometric fact we use is that, since r_m is a universal homeomorphism, pullback along it is an equivalence and its unit is an isomorphism. The rest is base change and composition of units, and each part uses the previous ones.

Proof of (1). The identities $p_m r_m = \Phi_X^m$ and $f'_m r_m = f$ hold by the definition of $r_m = (f, \Phi_X^m)$, since $f \Phi_X^m = \Phi_Y^m f$. The map r_m is the geometric base change of the relative q^m-power Frobenius of X_0/Y_0. It is integral, radicial and surjective [Stacks, 0CCB], and finite because it is integral and of finite type. For affine $Y_0 = \operatorname{Spec} R$ and $X_0 = \operatorname{Spec} S$, with F the q-power Frobenius of R, it is given on coordinate rings by

$$R \otimes_{F^m, R} S \longrightarrow S, \qquad b \otimes a \longmapsto f^{\#}(b) a^{q^m}.$$

By topological invariance for classical objects of finite Tor amplitude, r_m^* is an equivalence, so the unit η_{r_m} in the construction is an isomorphism. In the construction, the first arrow is base change for compact supports, and the last comes from the composition of direct images with compact support, since r_m is proper. Moreover, pasting the cartesian squares defining X'_m and X'_n, we get the comparison for $m + n$, because the relative Frobenii compose and the unit of their composite is the composite of the units. Thus

$$\theta_{f,m+n,K} = \theta_{f,m,(\Phi_X^n)^*K} \circ (\Phi_Y^m)^* \theta_{f,n,K},$$

which is the compatibility with iteration.

Proof of (2). The formula for ψ'_m follows by induction on m from the displayed iteration identity and the naturality of θ.

Proof of (3). Both endomorphisms are a pullback map followed by $R\Gamma_c$ of the correspondence, so we must see that the pullback along Φ_Y^m on $R\Gamma_c(Y, L)$ becomes the pullback along Φ_X^m on $R\Gamma_c(X, K)$. For a finite map h we also write h^* for the pullback map on cohomology with compact support, defined by the unit of $h^* \dashv Rh_*$. Since base change is compatible with these units, the pullback along Φ_Y^m, followed by the base change isomorphism, is the pullback along p_m, and the next arrow in $\theta_{f,m}$ becomes r_m^*. Their composite is $r_m^* p_m^* = (p_m r_m)^* = (\Phi_X^m)^*$, and composing with $R\Gamma_c(\psi_m)$ we see that the operators are intertwined.

Proof of (4). We pull the construction back to the fixed point y. Since Φ_Y^m fixes $\bar{y}$, the fibre of $X'_m \to Y$ over $\bar{y}$ is again the fibre $X_{\bar{y}}$ of f, and r_m restricts to a self-map of $X_{\bar{y}}$. For $f : \mathbf{A}^1 \to s$ and $m = 1$, for instance, $X'_1 = \mathbf{A}^1$ and r_1 is $t \mapsto t^q$. In general, the base change of r_m to this fibre is its q^m-power scheme Frobenius, and by proper base change its unit restricts to the unit on the fibre. Hence the same calculation identifies $\psi'_{m,\bar{y}}$ with the Frobenius operator on the cohomology with compact support of the fibre. Moreover, at a rational point x of the fibre, the restricted correspondence $\psi_m|_{X_{\bar{y}}}$ has stalk map $\psi_{m,\bar{x}}$, by composition of pullbacks. ∎

To reduce the trace formula to pieces of X_0, we cut X_0 into an open subscheme and its closed complement.

Lemma 12.19 (Additivity for an open subscheme and its complement)**.** *Let $j_0 : U_0 \hookrightarrow X_0$ be open with reduced closed complement $i_0 : Z_0 \to X_0$, and let $K \in D^b_{\mathrm{ctf}}(X, A)$. A Frobenius correspondence ψ on K induces Frobenius correspondences on j^*K, i^*K, $j_!j^*K$ and Ri_*i^*K, whose stalk maps are the restrictions of ψ_m. Writing T_{c_m} also for the endomorphisms of $R\Gamma_c$ induced by the Frobenius correspondences on U and Z, for every $m \geq 1$*

$$\mathrm{tr}_A(T_{c_m}(\psi_m) \mid R\Gamma_c(X, K)) = \mathrm{tr}_A(T_{c_m} \mid R\Gamma_c(U, j^*K)) + \mathrm{tr}_A(T_{c_m} \mid R\Gamma_c(Z, i^*K)).$$

Proof. The four induced correspondences come from base change, and $R\Gamma_c$ of the localization triangle is a two-step filtered object with an endomorphism, on which traces add.

Since Frobenius preserves opens defined over the finite field, its square with j is cartesian, and by base change we get the correspondences on j^*K and $j_!j^*K$. The Frobenius square of the closed complement, however, is cartesian only after passing to reductions, and we obtain the correspondences on i^*K and Ri_*i^*K from Lemma lemma-comparison-via-relative-frobenius–(2) and nilpotent invariance. We then apply the functorial localization triangle $j_!j^*K \to K \to Ri_*i^*K$ to the Frobenius correspondence and take direct images with compact support. Since the triangle is functorial, the Frobenius correspondence becomes an endomorphism of a two-step filtered object, and hence the traces add by Proposition 11.2*. Finally, Lemma lemma-comparison-via-relative-frobenius for j and i identifies the endomorphisms T_{c_m} of the cohomology of $j_!j^*K$ and Ri_*i^*K with those on U and Z. ∎

12.4 Finite coefficients

The proof uses a single compactification, $\mathbf{A}^1 \subset \mathbf{P}^1$, so the Lefschetz–Verdier formula and the curve computation of Proposition 12.16 are applied only on $\mathbf{P}^1$. This suffices because every affine

scheme of finite type over $\mathbf{F}_q$ is a closed subscheme of some $\mathbf{A}^d$, and the fibres of a coordinate projection lie in $\mathbf{A}^1$. On $\mathbf{P}^1$ the only new fixed point is the rational point ∞, where $j_!K$ has zero stalk. The fibres over points of $Y_0(\mathbf{F}_{q^m})$ are schemes over $\mathbf{F}_{q^m}$, so we prove the theorem for all finite fields at once. The other coefficient rings are treated in Section 12.5.

The base of the induction is the case of finite schemes. There we can see directly what the formula counts. For example, take $X_0 = \operatorname{Spec} \mathbf{F}_{q^2}$ and $K = A$. Then the scheme X consists of two points interchanged by Φ_X. Hence T_{c_1} is the swap of $A \oplus A$, with trace $0 = \#X_0(\mathbf{F}_q)$, and T_{c_2} is the identity, with trace $2 = \#X_0(\mathbf{F}_{q^2})$. In general T_{c_m} is a block permutation matrix, and only the points fixed by Φ_X^m contribute to its trace.

The iterates ψ_m, the endomorphisms $T_{c_m}(\psi_m)$ and the stalk maps $\psi_{m,\bar{x}}$ are those of Definition 12.2.

Theorem 12.20 (Trace formula for a Frobenius correspondence)**.** *Let $X_0/\mathbf{F}_q$ be separated and of finite type, put $X = X_0 \times_{\mathbf{F}_q} k$, let A be any finite commutative ℓ-primary ring and $K \in D^b_{\mathrm{ctf}}(X, A)$, and let $\psi : \Phi_X^* K \to K$ be any Frobenius correspondence on K. For every $m \geq 1$, in A,*

$$\operatorname{tr}_A(T_{c_m}(\psi_m) \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)) = \sum_{x \in X_0(\mathbf{F}_{q^m})} \operatorname{tr}_A(\psi_{m,\bar{x}} \mid K_{\bar{x}}).$$

Proof. Both sides are defined, because the complexes are perfect by Lemma 10.1*. We prove the assertion simultaneously for all finite fields, all Frobenius correspondences and all m, so that $\mathbf{F}_q$ may be replaced by $\mathbf{F}_{q^m}$ when we pass to a rational fibre. We go in four steps: finite schemes; the affine line, where we use the Lefschetz–Verdier formula on $\mathbf{P}^1$; closed subschemes of $\mathbf{A}^d$, by projecting away one coordinate; and arbitrary X_0, by cutting it into affine pieces.

Finite schemes. If X_0 is finite, both sides of the formula can be read off from a block matrix. Indeed, up to nilpotents, X is the finite disjoint union of its geometric points. By nilpotent invariance, $R\Gamma_c(X, K) = \bigoplus_{z \in X(k)} K_z$. In this decomposition, the pullback along Φ_X^m given by the unit permutes these summands, and the block of $T_{c_m}(\psi_m)$ from $K_{\Phi_X^m(z)}$ to K_z is the stalk of ψ_m at z, a map $K_{\Phi_X^m(z)} = ((\Phi_X^m)^* K)_z \to K_z$. On a finite projective model only the diagonal blocks contribute to the trace. These occur exactly at the points $z \in X_0(\mathbf{F}_{q^m})$, with the required stalk maps.

The affine line. This is the only step where we compute a local term. We pass to $\mathbf{P}^1$, where the Lefschetz–Verdier formula applies, and extend K by zero, so that the new fixed point ∞ contributes nothing. Let $j : \mathbf{A}^1_k \hookrightarrow \mathbf{P}^1_k$. The correspondence ψ_m extends to a correspondence $\bar{\psi}_m$ on $j_!K$, compatible with iteration. Indeed, j is defined over $\mathbf{F}_q$, so its square with the Frobenius maps is cartesian, and base change for this square gives $(\Phi^m_{\mathbf{P}^1})^* j_! K \simeq j_!(\Phi^m_{\mathbf{A}^1})^* K$. Lemma lemma-comparison-via-relative-frobenius, applied to j, shows that the composition isomorphism

$$R\Gamma_c(\mathbf{A}^1, K) \simeq R\Gamma(\mathbf{P}^1, j_! K)$$

intertwines the endomorphisms T_{c_m} induced by ψ and $\bar{\psi}$. The graph is proper on the proper curve $\mathbf{P}^1$, so by Theorem 11.13* the trace on the right is the sum of the local terms at the fixed points. By Proposition 12.16, each local term is a trace on a stalk. In particular, the local term at infinity vanishes, because the stalk of $j_!K$ there is zero. The other fixed points and stalk maps are those of $\mathbf{A}^1_{\mathbf{F}_q}$. This proves the formula for the affine line.

Closed subschemes of the affine line. Let $X_0 = \operatorname{Spec} \mathbf{F}_q[t]/I$. If $I = 0$, the previous paragraph applies. Otherwise I contains a nonzero polynomial, so X_0 is finite over $\mathbf{F}_q$ and the case of finite schemes applies. This also covers nonreduced subschemes.

Induction on affine coordinates. We push K forward along a coordinate projection. This does not change the global side. The local side over a point of the base becomes the global side of a fibre lying in $\mathbf{A}^1$. Suppose that the assertion holds for closed subschemes of $\mathbf{A}^{d-1}$ over every finite field, and let $X_0 \hookrightarrow \mathbf{A}^d_{\mathbf{F}_q}$ be closed. Let $f_0 : X_0 \to Y_0 = \mathbf{A}^{d-1}_{\mathbf{F}_q}$ forget the last coordinate, and let $f : X \to Y$ be its base change to k. The complex $L = Rf_!K$ lies in $D^b_{\mathrm{ctf}}(Y, A)$ by Proposition 7.1*, and we equip it with the Frobenius correspondence ψ' of Lemma lemma-comparison-via-relative-frobenius. As Lemma lemma-comparison-via-relative-frobenius identifies the endomorphisms T_{c_m}, the induction hypothesis shows that

$$\mathrm{tr}_A(T_{c_m}(\psi_m) \mid R\Gamma_c(X, K)) = \sum_{y \in Y_0(\mathbf{F}_{q^m})} \mathrm{tr}_A(\psi'_{m,\bar{y}} \mid L_{\bar{y}}).$$

By Lemma lemma-comparison-via-relative-frobenius, each summand is the trace on the cohomology with compact support of the fibre $X_{0,y}$, a closed subscheme of $\mathbf{A}^1_{\mathbf{F}_{q^m}}$. Here the endomorphism is given by the restriction of ψ_m to the fibre and the first power of the Frobenius over $\mathbf{F}_{q^m}$. The case of closed subschemes of the affine line then turns this summand into the sum of the traces on stalks over $X_{0,y}(\mathbf{F}_{q^m})$. As y varies, these sets partition $X_0(\mathbf{F}_{q^m})$, which proves the affine case.

General schemes. Both sides are additive for a decomposition into an open subscheme and its closed complement: the global side by Lemma 12.19, and the local side because the rational points split. It is therefore enough to cut X_0 into affine pieces. Choose a finite affine open cover $U_1, \dots, U_N$ of X_0, let $W_i = \bigcup_{j \le i} U_j$ and $V_i = W_i \setminus W_{i-1}$, with the reduced closed structure. Each V_i is closed in the affine U_i, hence affine, and it is the closed complement of W_{i-1} in W_i. Applying Lemma 12.19 successively, we see that the traces add over this stratification. The affine case applied to each V_i now gives the equality for all separated X_0 of finite type. ∎

12.5 ℓ-adic coefficients

To pass from finite coefficient rings to $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$ we follow the steps of Proposition 7.12*. Since the statement is now an equality of traces in Λ rather than an isomorphism, we carry the steps out directly. First, for $\mathcal{O}_E$ we reduce modulo π_E^n. This works because both sides are traces of perfect complexes, traces commute with reduction, and $\mathcal{O}_E$ embeds in $\varprojlim_n \mathcal{O}_E/\pi_E^n$. Next, for E we clear denominators: a Frobenius correspondence ψ_E with coefficients E need not come from an integral one; only some multiple $\pi_E^N \psi_E$ does, and this integral model need not be invertible even when ψ_E is (Lemma 12.22). This is why we prove the formula for correspondences that are not invertible. Finally, for $\overline{\mathbf{Q}}_\ell$ we descend to a finite extension of $\mathbf{Q}_\ell$.

The Frobenius comparison κ_m of Lemma 12.3 extends to the coefficient rings $\mathcal{O}_E$, E and $\overline{\mathbf{Q}}_\ell$. Let $K \in D^b_c(X, \mathcal{O}_E)$ have tower of reductions (K_n) modulo π_E^n (Definition 2.61*). Both $(\Phi_X^m)^*$ and Fr_X^{m*} preserve derived completeness and commute with reduction, so each is the derived limit of its values on the K_n. The finite comparisons κ_m come from a single functorial identification on the classical site. Hence they are compatible with reduction of coefficients, and their derived limit is the integral κ_m. The finite comparisons are compatible with iteration and with the units of the pullbacks, and therefore so is the integral one. It follows that the integral κ_m identifies the endomorphisms T_{c_m} with the geometric Frobenius operators as in Lemma 12.3. For E we localize on integral models. By the localized Hom comparison (9.21) the result does not depend on the choice of model. For $\overline{\mathbf{Q}}_\ell$ we descend from finite extensions.

Proposition 12.21 (Integral trace formula)**.** *Let $K \in D^b_c(X, \mathcal{O}_E)$ and let $\theta : \Phi_X^* K \to K$ be any morphism, that is, a Frobenius correspondence with integral coefficients. Define its iterates θ_m, the endomorphisms $T_{c_m}(\theta_m)$ and the stalk maps $\theta_{m,\bar{y}}$ by the formulas of Definition 12.2. Then for every $m \ge 1$ we have*

$$\operatorname{tr}_{\mathcal{O}_E}(T_{c_m}(\theta_m) \mid R\Gamma_c(X_{\text{pro\'et}}, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \operatorname{tr}_{\mathcal{O}_E}(\theta_{m,\bar{y}} \mid K_{\bar{y}}), \tag{12.1}$$

where both sides are categorical traces of perfect complexes over $\mathcal{O}_E$. Their reductions modulo π_E^n are the corresponding traces of Theorem 12.20 for K_n and the reduction $\bar{\theta}_n : \Phi_X^* K_n \to K_n$ of θ.

Proof. We show that both sides reduce modulo π_E^n to the two sides of the finite formula, by identifying the reductions of the complexes, of the operators and of the traces. This suffices, since an element of $\mathcal{O}_E$ that vanishes modulo every π_E^n is zero.

Complexes and operators. By Lemma 10.1*, $R\Gamma_c(X_{\text{pro\'et}}, K)$ and the geometric stalks are perfect, and Proposition 7.10* provides canonical isomorphisms

$$R\Gamma_c(X_{\text{pro\'et}}, K) \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n \simeq R\Gamma_c(X_{\text{pro\'et}}, K_n),$$
$$K_{\bar{y}} \otimes^{\mathbf{L}}_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n \simeq (K_n)_{\bar{y}}.$$

Under these isomorphisms the reductions of the integral operators are the operators of $\bar{\theta}_n$, because extension of scalars commutes with the pullbacks, with the units defining the endomorphisms T_{c_m}, and with composition, and hence with the iterates θ_m. Moreover Theorem 12.20 applies for every n, because $K_n \in D^b_{\text{ctf}}(X, \mathcal{O}_E/\pi_E^n)$.

Traces. Represent each of the two integral complexes by a bounded complex $P^\bullet$ of finite projective $\mathcal{O}_E$-modules and its endomorphism by a chain map $v^\bullet$. By Proposition 11.2* its trace is

$$\operatorname{tr}_{\mathcal{O}_E}(v) = \sum_i (-1)^i \operatorname{Tr}(v^i \mid P^i).$$

The reduction of this expression modulo π_E^n is the trace on $P^\bullet \otimes_{\mathcal{O}_E} \mathcal{O}_E/\pi_E^n$, as each P^i is finite projective.

Conclusion. The reduction of each side of (12.1) is therefore the corresponding side of the finite formula for $(K_n, \bar{\theta}_n)$. Hence the difference of the two sides lies in $\pi_E^n \mathcal{O}_E$ for every n. It is zero because $\mathcal{O}_E \hookrightarrow \varprojlim_n \mathcal{O}_E/\pi_E^n$ is injective. ∎

To pass from $\mathcal{O}_E$ to E we need an integral Frobenius correspondence whose rationalization is the given one, up to a power of π_E. The next lemma provides it and records how the operators of both sides of the trace formula change under this rescaling.

Lemma 12.22 (Integral models for rational Frobenius correspondences)**.** *Let $K_E \in D^b_c(X, E)$ and let $\psi_E : \Phi_X^* K_E \to K_E$ be any morphism. There exist $K_0 \in D^b_c(X, \mathcal{O}_E)$, an isomorphism $K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_E} E \simeq K_E$, an integer $N \geq 0$, and a Frobenius correspondence $\theta : \Phi_X^* K_0 \to K_0$ such that*

$$\psi_E = \pi_E^{-N}(\theta \otimes^{\mathbf{L}}_{\mathcal{O}_E} E).$$

For every $m \geq 1$ the operators of Definition 12.2 satisfy

$$\begin{aligned} T_{c_m}((\psi_E)_m) &= \pi_E^{-Nm}(T_{c_m}(\theta_m) \otimes^{\mathbf{L}}_{\mathcal{O}_E} E), \\ (\psi_E)_{m,\bar{y}} &= \pi_E^{-Nm}(\theta_{m,\bar{y}} \otimes^{\mathbf{L}}_{\mathcal{O}_E} E). \end{aligned} \tag{12.2}$$

Proof. Morphisms in the rational category are integral morphisms divided by a power of π_E. This gives K_0 and θ. The formulas for the operators hold because the operators of Definition 12.2 commute with extension of scalars.

By Proposition 3.27*, extension of scalars induces an equivalence

$$D_c^b(X, \mathcal{O}_E)[1/\ell] \simeq D_c^b(X, E),$$

where the source has the objects of $D_c^b(X, \mathcal{O}_E)$ and Hom groups tensored with E. By essential surjectivity, K_E has an integral model K_0. The model constructed there is built by finite cones and shifts from the integral generators of Lemma lemma-integral-models, and hence in general it is not a lattice in a lisse sheaf.

Since inverse image preserves integral constructibility, $\Phi_X^* K_0$ is an integral model of $\Phi_X^* K_E$. By the description of the Hom groups above, applied to the pair $(\Phi_X^* K_0, K_0)$, we may write $\psi_E = \pi_E^{-N}(\theta \otimes^{\mathbf{L}}_{\mathcal{O}_E} E)$. Since pullback is $\mathcal{O}_E$-linear, each of the m factors of the iterate contributes π_E^{-N}, and so $(\psi_E)_m = \pi_E^{-Nm}(\theta_m \otimes^{\mathbf{L}} E)$. By Proposition 7.10*, the change-of-coefficients maps for cohomology with compact support and for stalks respect the units of the pullback adjunctions. Hence the pullback map $(\Phi_X^m)^*$ on $R\Gamma_c$ for E is the extension of scalars of the integral one. The same holds for the identification $((\Phi_X^m)^* K_E)_{\bar{y}} \simeq (K_E)_{\bar{y}}$ at a fixed point $\bar{y}$ of Φ_X^m, because the comparison for inverse images is compatible with composition and with extension of scalars. Applying T_{c_m} and taking stalks at $\bar{y}$ in the formula for $(\psi_E)_m$ gives (12.2). ∎

Remark 12.23 (Integral Frobenius correspondences need not be invertible)**.** In Lemma 12.22, K_0 need not be a lattice in a lisse sheaf, and θ need not be invertible even when ψ_E is. The factor π_E^{-N} is needed because ψ_E itself need not be integral: on a point, the automorphism ℓ^{-1} of E preserves no lattice. Even with this factor, θ cannot always be chosen invertible, already on a point. On E^2, the two eigenvalues of $\mathrm{diag}(1, \ell)$ cannot both become units after multiplying the operator by a single scalar. So Lemma 12.22 only makes $\pi_E^N \psi_E$ integral, and the reductions of θ may fail to be invertible. For this reason Theorem 12.20 is stated for arbitrary Frobenius correspondences.

With the integral formula and integral models of rational Frobenius correspondences, we obtain the trace formula stated at the beginning of this chapter for all coefficient rings.

Theorem 12.24 (Grothendieck–Lefschetz trace formula)**.** *Let X_0 be separated and of finite type over $\mathbf{F}_q$, put $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and suppose that $\ell \nmid q$. Let*

$$\Lambda \in \{A, \mathbf{Z}_\ell, \mathcal{O}_E, \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}, \qquad K \in D^b_{\mathrm{ctf}}(X, \Lambda),$$

with a Weil structure as in Definition 10.18. Here A is any finite commutative ℓ-primary ring, and for integral and rational coefficients, $D^b_{\mathrm{ctf}}(X, \Lambda) = D^b_c(X, \Lambda)$. Let $\mathrm{Frob}_{X,K}$ be the geometric Frobenius on $R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)$, and let $\mathrm{Frob}_{y,m}$ be the Frobenius acting on the stalk $K_{\bar{y}}$ at $y \in X_0(\mathbf{F}_{q^m})$. Then for every $m \geq 1$ we have, in Λ,*

$$\mathrm{tr}_\Lambda\big(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\mathrm{pro\acute{e}t}}, K)\big) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \mathrm{tr}_\Lambda\big(\mathrm{Frob}_{y,m} \mid K_{\bar{y}}\big). \tag{12.3}$$

The complexes inside the traces are perfect over Λ, and the formula is compatible with the change-of-coefficients maps of Proposition 7.10. For $\Lambda = \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$, it reads*

$$\sum_i (-1)^i \operatorname{Tr}(\mathrm{Frob}_{X,K}^m \mid H_c^i(X_{\mathrm{pro\acute{e}t}}, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} \sum_j (-1)^j \operatorname{Tr}(\mathrm{Frob}_{y,m} \mid H^j(K_{\bar{y}})), \tag{12.4}$$

where all sums over degrees are finite.

For finite and integral rings, (12.3) is an identity of categorical traces of perfect complexes, which makes sense even when the cohomology modules are not projective.

Proof of Theorem 12.24. In each case we turn the Weil structure into a Frobenius correspondence by composing with κ_1, and we use Lemma 12.3 to see that this does not change the operators. We then treat the coefficient rings in the order finite, integral, rational, algebraic, each case using the previous one.

For finite A, we apply Theorem 12.20 to $\psi = \phi\kappa_1$. Then Lemma 12.3 identifies $T_{c_m}(\psi_m)$ with $\mathrm{Frob}_{X,K}^m$ and, at each fixed point, $\psi_{m,\bar{y}}$ with $\mathrm{Frob}_{y,m}$. Similarly, for $\mathcal{O}_E$, composing the Weil structure with κ_1 we get a Frobenius correspondence $\Phi_X^* K \to K$. By the integral form of Lemma 12.3, the endomorphisms it induces on $R\Gamma_c$ and on the stalks are those induced by the Weil structure. The identity therefore follows from Proposition 12.21. In particular, taking $E = \mathbf{Q}_\ell$ gives the case of $\mathbf{Z}_\ell$.

For coefficients E, the Weil structure composed with κ_1 is a Frobenius correspondence ψ_E, and by the rational form of Lemma 12.3 the endomorphisms it induces on $R\Gamma_c$ and on the stalks are those induced by the Weil structure. We apply Lemma 12.22 to ψ_E, apply (12.1) to (K_0, θ), extend scalars to E, and multiply by π_E^{-Nm}. By linearity of categorical traces and their compatibility with extension of scalars (Proposition 11.2*), together with (12.2), this gives the identity for every m. The case $E = \mathbf{Q}_\ell$ gives the case of $\mathbf{Q}_\ell$.

For $\overline{\mathbf{Q}}_\ell$, complexes and morphisms are colimits of their versions over finite extensions, so it is enough to find the Weil complex over some finite E and to check that the answer does not depend on the choice. By Proposition 7.11* the complex K, its Weil structure ϕ and ϕ^{-1} descend to a common finite extension $E/\mathbf{Q}_\ell$. After a further finite extension the two identities saying that these maps are inverse to each other hold as well. Thus we get a Weil complex at a finite stage, and by extension of scalars its formula over E gives (12.3). If we choose another representative, any isomorphism compatible with the Weil structures descends to a common finite extension, so the resulting operators are conjugate. Since the trace is compatible with extension of scalars and invariant under conjugation, the result does not depend on the stage or on the representative.

The complexes are perfect by Lemma 10.1*. Compatibility with the change-of-coefficients maps holds because the pullbacks, the endomorphisms T_{c_m} and the iterated Frobenius correspondences commute with the comparisons of Proposition 7.10*. Finally, over a field the categorical trace is the alternating trace on cohomology by Proposition 11.2*, and this proves (12.4). ∎

12.6 The function–sheaf dictionary

Under the trace formula, $Rf_!$ becomes summation over fibres and $\otimes$ becomes multiplication of functions on rational points. This is Grothendieck's function–sheaf dictionary [SGA4½, Sommes trig.] (Section 1.3*), and in Part III we compute with sheaves through their functions in this way.

Definition 12.25 (Trace function)**.** Let (K, ϕ) be a Weil complex on X as in Theorem 12.24, and let $m \geq 1$. The *trace function* of K is

$$t_{K,m} : X_0(\mathbf{F}_{q^m}) \longrightarrow \Lambda, \qquad t_{K,m}(x) = \operatorname{tr}_\Lambda(\mathrm{Frob}_{x,m} \mid K_{\bar{x}}).$$

For Λ a field, $t_{K,m}(x)$ is the alternating sum of the traces of Frobenius on the stalks $\mathcal{H}^i(K)_{\bar{x}}$ (Proposition 11.2*). For example, the constant sheaf Λ_X with its constant Weil structure has trace function 1. In terms of trace functions, the theorem says that $\mathrm{tr}_\Lambda(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\text{pro}\acute{e}t}, K))$ is the sum of $t_{K,m}$ over $X_0(\mathbf{F}_{q^m})$. Applied to the fibres of a morphism, it shows that direct image with compact support sums trace functions over the fibres.

Corollary 12.26 (Trace functions and direct image with compact support)**.** *Let $f_0 : X_0 \to Y_0$ be a morphism of separated $\mathbf{F}_q$-schemes of finite type, with geometric base change $f : X \to Y$, and let (K, ϕ) be a Weil complex on X with $K \in D^b_{\mathrm{ctf}}(X, \Lambda)$ for an admissible Λ. Give $Rf_!K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$ the Weil structure induced by ϕ (after Definition 10.18*). Then for every $m \geq 1$ and $y \in Y_0(\mathbf{F}_{q^m})$,*

$$t_{Rf_!K,m}(y) = \sum_{\substack{x \in X_0(\mathbf{F}_{q^m}) \\ f_0(x) = y}} t_{K,m}(x).$$

Proof. The left side is a trace on a stalk of $Rf_!K$. By base change that stalk is the compactly supported cohomology of a fibre. Hence the corollary is the trace formula for the fibre, once we check that the Frobenius operators correspond.

Fix y and let $X_{0,y}$ be the fibre of f_0 over y. It is a separated scheme of finite type over $\mathbf{F}_{q^m}$, and its rational points are the $x \in X_0(\mathbf{F}_{q^m})$ with $f_0(x) = y$. Restrict K to its geometric fibre $X_{\bar{y}}$ and give the restriction the Weil structure over $\mathbf{F}_{q^m}$ obtained from the m-th iterate of ϕ. By base change for compact supports, $(Rf_!K)_{\bar{y}}$ is $R\Gamma_c(X_{\bar{y}}, K|_{X_{\bar{y}}})$, and $\mathrm{Frob}_{y,m}$ is the geometric Frobenius operator of this Weil complex. For finite A this is part (4) of Lemma 12.18, applied to the Frobenius correspondence $\phi\kappa_1$ and translated by Lemma 12.3. By the comparisons of Proposition 7.10* it extends to the other coefficient rings. Since the Frobenius acting on the stalk of the Weil complex on the fibre at a rational point x of $X_{0,y}$ is $\mathrm{Frob}_{x,m}$, the stated identity is Theorem 12.24 for $X_{0,y}$ over $\mathbf{F}_{q^m}$, with exponent 1. ∎

Example 12.27 (The squaring map)**.** Let q be odd, let $f_0 : \mathbf{A}^1_{\mathbf{F}_q} \to \mathbf{A}^1_{\mathbf{F}_q}$ be $x \mapsto x^2$, and let $K = \Lambda_X$ with its constant Weil structure, so that $t_{K,m} = 1$. The fibre of f_0 over $y \in \mathbf{F}_{q^m}$ has $1 + \chi_m(y)$ rational points, where χ_m is the quadratic character of $\mathbf{F}^*_{q^m}$, extended by $\chi_m(0) = 0$. Hence

$$t_{Rf_!\Lambda,m}(y) = 1 + \chi_m(y).$$

The trace function of $Rf_!\Lambda$ is not constant, although the sheaf we pushed forward is: over $\mathbf{G}_m$ the map is a finite étale double cover, and $\chi_m(y)$ records whether the fibre over y splits into two rational points or is a single point of degree 2. Summing over y recovers the trace formula on the source. Since χ_m takes the values ± 1 equally often on $\mathbf{F}^*_{q^m}$, the sum is $q^m = \#\mathbf{A}^1(\mathbf{F}_{q^m})$.

The other operations on functions also have counterparts on sheaves. These counterparts, together with the corollary, are the entries of the dictionary that we use. In this dictionary, morphisms come from morphisms of separated $\mathbf{F}_q$-schemes of finite type, and the operations carry the Weil structures described after Definition 10.18*.

Proposition 12.28 (The function–sheaf dictionary)**.** *Let Λ be as in Theorem 12.24, let $m \geq 1$, and let K, L be Weil complexes in $D^b_{\mathrm{ctf}}(X, \Lambda)$.*

1. *For $g_0 : W_0 \to X_0$ and $w \in W_0(\mathbf{F}_{q^m})$, $t_{g^*K,m}(w) = t_{K,m}(g_0(w))$.*
2. *$t_{K \otimes^{\mathbf{L}}_\Lambda L,m} = t_{K,m}\, t_{L,m}$.*

3. *For $f_0 : X_0 \to Y_0$ and $y \in Y_0(\mathbf{F}_{q^m})$, $t_{Rf_!K,m}(y) = \sum_{x \in X_0(\mathbf{F}_{q^m}),\, f_0(x)=y} t_{K,m}(x)$.*
4. *For $U_0 \subset X_0$ open with closed complement Z_0, and $j : U \to X$, $i : Z \to X$ the inclusions, $t_{K,m} = t_{j_!j^*K,m} + t_{i_*i^*K,m}$, where $t_{j_!j^*K,m}$ is $t_{K,m}$ on $U_0(\mathbf{F}_{q^m})$ and 0 on $Z_0(\mathbf{F}_{q^m})$.*
5. *$\sum_{x \in X_0(\mathbf{F}_{q^m})} t_{K,m}(x) = \operatorname{tr}_\Lambda(\mathrm{Frob}_{X,K}^m \mid R\Gamma_c(X_{\text{proét}}, K))$.*

Proof. Part (5) is Theorem 12.24 and part (3) is Corollary 12.26.

For (1), the stalk of g^*K at $\bar{w}$ is $K_{\bar{x}}$ with $x = g_0(w)$, and the Weil structure of g^*K is $g^*\phi$ composed with the base change isomorphism. Under this identification of stalks the m-th iterate of the Weil structure of g^*K at $\bar{w}$ is the m-th iterate of ϕ at $\bar{x}$, so the two Frobenius endomorphisms agree.

For (2), the stalk of $K \otimes^{\mathbf{L}}_\Lambda L$ at $\bar{x}$ is $K_{\bar{x}} \otimes^{\mathbf{L}}_\Lambda L_{\bar{x}}$, since pullback to $\bar{x}$ commutes with derived tensor products, and its Frobenius is the tensor product of the two Frobenius endomorphisms. These stalks are perfect (Lemma 10.1*), so we may represent them by bounded complexes P, Q of finite projective modules and the endomorphisms by chain maps v, w. Now the term of degree n of $P \otimes Q$ is $\bigoplus_{i+j=n} P^i \otimes Q^j$, the trace of $v^i \otimes w^j$ on $P^i \otimes Q^j$ is $\operatorname{Tr}(v^i)\operatorname{Tr}(w^j)$, and $(-1)^n = (-1)^i(-1)^j$. By the chain formula of Proposition 11.2* the trace on $P \otimes Q$ is therefore the product of the traces on P and on Q.

For (4), note that $j_! = Rj_!$ and $i_* = Ri_!$, with the same Weil structures. By (3), $t_{j_!j^*K,m}(x)$ is the sum of $t_{j^*K,m}$ over the fibre of j_0 at x. This fibre is the point x if $x \in U_0$, and it is empty otherwise. By (1), $t_{j^*K,m}(x) = t_{K,m}(x)$ for $x \in U_0(\mathbf{F}_{q^m})$. The same argument applies to i_*i^*K, and the two functions add up to $t_{K,m}$. ∎

The Fourier transform of Chapter 16* is one more entry, and the proposition shows why it corresponds to the discrete transform. Let p_1, p_2 be the projections of $\mathbf{A}^1 \times \mathbf{A}^1$, let $\mu(x,y) = xy$, let $\psi : \mathbf{F}_q \to E^\times$ be a nontrivial additive character, and let $\mathcal{K}_\psi = \mu^*\mathcal{L}_\psi$, where $\mathcal{L}_\psi$ is the Artin–Schreier sheaf of Lemma 16.1* (proved in Chapter 16* below). Here ψ is a character, not a Frobenius correspondence. With this kernel the transform is $\mathrm{FT}_\psi K = Rp_{2!}(p_1^*K \otimes \mathcal{K}_\psi)[1]$. For $y \in \mathbf{F}_{q^m}$ we compute

$$\begin{aligned}
t_{\mathrm{FT}_\psi K,m}(y) &= -\,t_{Rp_{2!}(p_1^*K\otimes\mathcal{K}_\psi),m}(y) && \text{(shift)}\\
&= -\sum_{x\in\mathbf{F}_{q^m}} t_{p_1^*K\otimes\mathcal{K}_\psi,m}(x,y) && \text{by (3)}\\
&= -\sum_{x\in\mathbf{F}_{q^m}} t_{p_1^*K,m}(x,y)\, t_{\mu^*\mathcal{L}_\psi,m}(x,y) && \text{by (2)}\\
&= -\sum_{x\in\mathbf{F}_{q^m}} t_{K,m}(x)\,\psi(\mathrm{Tr}_{\mathbf{F}_{q^m}/\mathbf{F}_q}(xy)) && \text{by (1) and Lemma 16.1.1.}
\end{aligned}$$

The first equality holds because a shift changes the sign of a trace, by the chain formula of Proposition 11.2*. Each of the other three steps is one operation in the formula $\hat{t}(y) = \sum_x t(x)\psi(xy)$: pull t back to the plane, multiply by the kernel, and sum over the fibres of p_2.

12.7 Rationality and the functional equation

Over a field of characteristic zero, the traces of all powers of an endomorphism determine its characteristic polynomial. Taking the trace formula for all $m \geq 1$, we therefore obtain an identity of power series: a product over closed points equals an alternating product of characteristic polynomials of Frobenius on cohomology. This step divides by m, so in the determinant formulas below Λ is one of the fields $\mathbf{Q}_\ell$, E, $\overline{\mathbf{Q}}_\ell$. Characteristic zero is needed: on $\mathbf{F}_p^{\oplus p}$ every power of the

identity has trace $p = 0$, as for the zero endomorphism, but $\det(1 - T \cdot \mathrm{id}) = (1-T)^p = 1 - T^p$ is not 1.

For $K = \Lambda_X$ the product over cohomology is finite, so $Z(X_0, T)$ is rational, and for smooth proper X_0 duality pairs the factors in degrees i and $2d - i$ (Corollary 12.33).

Lemma 12.29 (Logarithm of a determinant)**.** *Let L be a field of characteristic zero. For an endomorphism v of a finite-dimensional L-vector space V we have*

$$\det(1 - Tv \mid V)^{-1} = \exp\left(\sum_{m \geq 1} \mathrm{Tr}(v^m \mid V)\frac{T^m}{m}\right) \quad \text{in } 1 + TL[[T]]. \tag{12.5}$$

For an endomorphism v of a perfect complex P in $D(L)$, put

$$\det(1 - Tv \mid P) := \prod_i \det(1 - Tv \mid H^i(P))^{(-1)^i}.$$

Then (12.5) holds for P, with $\mathrm{Tr}(v^m \mid V)$ replaced by the categorical trace $\mathrm{tr}_L(v^m \mid P)$.

Proof. For a single eigenvalue α the identity is $-\log(1 - \alpha T) = \sum_m \alpha^m T^m/m$, and the general case could be reduced to this by triangularizing v over an algebraic closure. Instead we use logarithmic derivatives, which allows us to stay over L. In $L[[T]]$, the matrix $1 - Tv$ is invertible, with inverse $\sum_{r \geq 0} T^r v^r$. Moreover, the logarithmic derivative of the determinant of an invertible matrix M over $L[[T]]$ is $\mathrm{Tr}(M^{-1}M')$, and hence

$$\frac{d}{dT}\log\det(1 - Tv) = -\mathrm{Tr}\big(v(1 - Tv)^{-1}\big) = -\sum_{m \geq 1} \mathrm{Tr}(v^m)T^{m-1}.$$

Since both logarithms have constant coefficient zero and L has characteristic zero, we may integrate and exponentiate, and we obtain (12.5). For a perfect complex, the categorical trace over a field is the alternating sum of the traces on the finitely many nonzero cohomology spaces (Proposition 11.2*), and the assertion follows by multiplying the identities for the $H^i(P)$. ∎

The local factors of the Euler product are indexed by closed points. A closed point $x \in |X_0|$ of degree d over $\mathbf{F}_q$ corresponds to a Frobenius orbit of d geometric points, and its Frobenius on the stalk is the Frobenius going once around this orbit.

Theorem 12.30 (The trace formula as an Euler product)**.** *Let X_0 be separated and of finite type over $\mathbf{F}_q$, let $\Lambda \in \{\mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell\}$, and let $K \in D_c^b(X, \Lambda)$ carry a Weil structure, with stalk Frobenius maps $\mathrm{Frob}_{y,m}$ as in Theorem 12.24. For each closed point $x \in |X_0|$, let $\deg(x) = [\kappa(x) : \mathbf{F}_q]$ be its degree, choose one of the $\deg(x)$ points $y \in X_0(\mathbf{F}_{q^{\deg(x)}})$ lying over x, write $\bar{x} = \bar{y}$ for the corresponding geometric point, and let $\mathrm{Frob}_x = \mathrm{Frob}_{y,\deg(x)}$ be its action on $K_{\bar{x}}$. Another choice of y replaces Frob_x by a conjugate. Define*

$$L(X_0, K, T) := \prod_{x \in |X_0|} \det\big(1 - T^{\deg(x)}\,\mathrm{Frob}_x \mid K_{\bar{x}}\big)^{-1}. \tag{12.6}$$

For every n, only finitely many factors differ from 1 modulo T^n, so the product converges T-adically in $1 + T\Lambda[[T]]$, and

$$L(X_0, K, T) = \prod_i \det\bigl(1 - T \operatorname{Frob}_{X,K} \mid H^i_c(X_{\text{pro\'et}}, K)\bigr)^{(-1)^{i+1}}. \tag{12.7}$$

In particular, $L(X_0, K, T)$ is the expansion at zero of a rational function in $\Lambda(T)$ with value 1 at zero. If X_0 is proper, we may replace H^i_c by H^i.

Proof. Both sides are of the form exp of a power series, and we compare the logarithms coefficient by coefficient. The coefficient of T^m on the cohomological side is the global trace of Frob^m divided by m, and on the side of closed points it is the sum of the local traces over $X_0(\mathbf{F}_{q^m})$ divided by m. Hence the theorem is the trace formula for all m at once.

For convergence, note that X_0 has only finitely many closed points of degree at most any given bound n, because each of them lies under a point of the finite set $X_0(\mathbf{F}_{q^{n!}})$. The factor at x in (12.6) is 1 modulo $T^{\deg(x)}$, so the product converges.

Before comparing logarithms we rewrite the sum over points in Theorem 12.24 in terms of closed points. A closed point of degree d has exactly d points over $\mathbf{F}_{q^m}$ when $d \mid m$, and none otherwise. For instance, $\operatorname{Spec} \mathbf{F}_{q^2}$ has one closed point, of degree 2, with no $\mathbf{F}_q$-points and two $\mathbf{F}_{q^2}$-points. These d points form one Frobenius orbit, and the Weil structure gives isomorphisms between the stalks at consecutive points of the orbit. The Frobenius on the stalk at a point of the orbit is then the composite of these isomorphisms taken once around the orbit, starting at that point. Moving the starting point cyclically therefore conjugates this composite by one of these isomorphisms. Hence the operators at the $\deg(x)$ points are conjugate. In the same way, iterating the Weil structure shows that for $\deg(x) \mid m$ the operator $\mathrm{Frob}_{y,m}$ at each of the $\deg(x)$ points $y \in X_0(\mathbf{F}_{q^m})$ over x is conjugate to $\mathrm{Frob}_x^{m/\deg(x)}$. Since traces are invariant under conjugation, (12.3) reads

$$\operatorname{tr}_\Lambda(\mathrm{Frob}^m_{X,K} \mid R\Gamma_c(X_{\text{pro\'et}}, K)) = \sum_{\substack{x \in |X_0| \\ \deg(x) \mid m}} \deg(x)\, \operatorname{tr}_\Lambda(\mathrm{Frob}_x^{m/\deg(x)} \mid K_{\bar{x}}). \tag{12.8}$$

We multiply (12.8) by T^m/m and sum over $m \geq 1$. By Lemma 12.29, the left side becomes the logarithm of the right side of (12.7). In the sum over closed points on the right, we substitute $m = r \deg(x)$. Then the multiplicity $\deg(x)$ cancels the factor $\deg(x)$ in m, and Lemma 12.29 applied to Frob_x on $K_{\bar{x}}$, with $T^{\deg(x)}$ in place of T, shows that

$$\sum_{x \in |X_0|} \sum_{r \geq 1} \operatorname{tr}_\Lambda(\mathrm{Frob}_x^r \mid K_{\bar{x}}) \frac{T^{r \deg(x)}}{r} = \log L(X_0, K, T).$$

Each coefficient of T receives only finitely many contributions, so we may regroup the sums. Exponentiating proves (12.7). Rationality holds because the cohomological product has finitely many factors, each a polynomial or the reciprocal of one, with value 1 at zero. If X_0 is proper, compact and ordinary cohomology agree. ∎

For the constant sheaf every local factor is $(1 - T^{\deg(x)})^{-1}$, and the Euler product becomes the zeta function of X_0. This proves rationality of $Z(X_0, T)$ over $\mathbf{Q}_\ell$. A short argument with linear recurrences descends it to $\mathbf{Q}$.

Corollary 12.31 (Point counting and the zeta function)**.** *Give Λ_X the Weil structure of constant sheaves defined after Definition 10.18*. For every coefficient ring Λ of Theorem 12.24 and every $m \geq 1$ we have*

$$\operatorname{tr}_\Lambda(\mathrm{Frob}_{X,\Lambda_X}^m \mid R\Gamma_c(X_{\text{pro\'et}}, \Lambda_X)) = \#X_0(\mathbf{F}_{q^m}) \cdot 1_\Lambda. \tag{12.9}$$

For $\Lambda = \mathbf{Q}_\ell, E, \overline{\mathbf{Q}}_\ell$, this gives

$$\#X_0(\mathbf{F}_{q^m}) = \sum_i (-1)^i \operatorname{Tr}(\mathrm{Frob}_{X,\Lambda_X}^m \mid H_c^i(X_{\text{pro\'et}}, \Lambda_X)),$$

and

$$\begin{aligned} Z(X_0, T) &:= \exp\left(\sum_{m \geq 1} \#X_0(\mathbf{F}_{q^m}) \frac{T^m}{m}\right) \\ &= \prod_{x \in |X_0|} (1 - T^{\deg(x)})^{-1} = \prod_i \det(1 - T \operatorname{Frob}_{X,\Lambda_X} \mid H_c^i(X_{\text{pro\'et}}, \Lambda_X))^{(-1)^{i+1}}. \end{aligned}$$

The first two series lie in $1 + T\mathbf{Z}[[T]]$, and the last equality holds in $\Lambda[[T]]$. Moreover, $Z(X_0, T)$ belongs to $\mathbf{Q}(T)$.

Proof. The identities over Λ are the trace formula and the Euler product for the constant sheaf. What needs an argument is that $Z(X_0, T)$ has rational coefficients, which we deduce at the end from the rationality over $\mathbf{Q}_\ell$ by linear algebra.

For $K = \Lambda_X$, every stalk is Λ, and every stalk Frobenius $\mathrm{Frob}_{y,m}$ is its identity, with categorical trace 1_Λ. Thus (12.9) follows from Theorem 12.24, and the formula over the fields follows from its cohomological form (12.4). By the orbit count in the proof of Theorem 12.30, we have $\#X_0(\mathbf{F}_{q^m}) = \sum_{\deg(x) \mid m} \deg(x)$. Expanding each $(1 - T^{\deg(x)})^{-1}$ as a geometric series shows that the product over closed points has integer coefficients. By the orbit count, its logarithm is the series defining $Z(X_0, T)$. The last equality is Theorem 12.30 for $K = \Lambda_X$.

Rationality descends from $\mathbf{Q}_\ell$ to $\mathbf{Q}$ for the following reason. A power series is rational exactly when its coefficients eventually satisfy a linear recurrence with constant coefficients. The existence of such a recurrence is a system of linear equations, and a system with rational coefficients has a rational solution as soon as it has any solution. More precisely, write $Z(X_0, T) = \sum_{n \geq 0} z_n T^n$, with $z_n \in \mathbf{Z}$. Rationality over $\mathbf{Q}_\ell$ means that there are integers d, N and $b_1, \ldots, b_d \in \overline{\mathbf{Q}}_\ell$ such that $z_n + b_1 z_{n-1} + \cdots + b_d z_{n-d} = 0$ for all $n > N$. These are linear equations with rational coefficients in the d unknowns b_i. Their coefficient rows span a finite-dimensional subspace of $\mathbf{Q}^{d+1}$, and hence finitely many of the equations imply all the others. This finite rational linear system has a solution over $\mathbf{Q}_\ell$, and hence one over $\mathbf{Q}$ by Gaussian elimination. With these rational b_i, the product of $1 + b_1 T + \cdots + b_d T^d \in \mathbf{Q}[T]$ and $Z(X_0, T)$ is a polynomial. Therefore $Z(X_0, T) \in \mathbf{Q}(T)$. ∎

For example, let $X_0 = \mathbf{A}^1_{\mathbf{F}_q}$. Then $\#X_0(\mathbf{F}_{q^m}) = q^m$, so that $Z(X_0, T) = (1 - qT)^{-1}$. On the cohomological side, $H_c^i(X, \mathbf{Q}_\ell)$ vanishes except for $i = 2$, and $H_c^2(X, \mathbf{Q}_\ell) \simeq \mathbf{Q}_\ell(-1)$ is a line on which geometric Frobenius acts by q. The only factor is therefore $(1 - qT)^{-1}$, as it should be.

For $X_0 = \mathbf{P}^1_{\mathbf{F}_q}$ the cohomology is $\mathbf{Q}_\ell$ in degree 0 and $\mathbf{Q}_\ell(-1)$ in degree 2, so $Z(\mathbf{P}^1, T) = 1/((1 - T)(1 - qT))$. This matches $\#\mathbf{P}^1(\mathbf{F}_{q^m}) = 1 + q^m$. For a curve of positive genus the middle cohomology contributes as well, and for the running example a single point count determines it.

Example 12.32 (The running example: the trace formula)**.** For the elliptic curve $C_0 \colon y^2 = x^3 - x$ over $\mathbf{F}_3$, geometric Frobenius acts on $H^0(C, \mathbf{Q}_\ell) = \mathbf{Q}_\ell$ by 1 and on $H^2(C, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-1)$ by 3, and $H^1(C, \mathbf{Q}_\ell)$ is a plane (Example 10.23*). The point-counting formula reads

$$N_m = 1 - \operatorname{Tr}(F^m \mid H^1(C, \mathbf{Q}_\ell)) + 3^m.$$

From $N_1 = 4$ we get $\operatorname{Tr}(F \mid H^1) = 0$. Duality gives $\det(F \mid H^1) = 3$, so $\det(1 - TF \mid H^1) = 1 + 3T^2$. Hence

$$Z(C_0, T) = \frac{1 + 3T^2}{(1 - T)(1 - 3T)}.$$

The formula predicts $N_2 = 1 + 9 + 6 = 16$ and $N_3 = 1 + 27 - 0 = 28$, and direct counts over $\mathbf{F}_9$ and $\mathbf{F}_{27}$ confirm this. In general, for a genus-one curve over $\mathbf{F}_q$ the single number N_1 determines $P_1 = 1 - (1 + q - N_1)T + qT^2$, and therefore every N_m.

For smooth proper X_0, Poincaré duality relates the factors of the determinant formula in complementary degrees, and this gives a functional equation.

Corollary 12.33 (Functional equation)**.** *Let $X_0/\mathbf{F}_q$ be smooth and proper of pure dimension d, and let $\ell \nmid q$. Put $H^i = H^i(X_{\text{pro\'et}}, \mathbf{Q}_\ell)$, let F_i be the Frobenius operator F on H^i (Section 10.5*), and put*

$$b_i = \dim_{\mathbf{Q}_\ell} H^i, \qquad \chi = \sum_i (-1)^i b_i, \qquad D = \prod_i \det(F_i)^{(-1)^i}.$$

Then $b_{2d-i} = b_i$, $d\chi$ is even, $D^2 = q^{d\chi}$, and in $\mathbf{Q}_\ell(T)$

$$Z\Big(X_0, \frac{1}{q^d T}\Big) = (-1)^\chi D\, T^\chi\, Z(X_0, T) = \varepsilon q^{d\chi/2} T^\chi Z(X_0, T), \qquad \varepsilon = (-1)^\chi D q^{-d\chi/2} \in \{1, -1\}.$$

Proof. Duality gives the relations $b_{2d-i} = b_i$ and $D^2 = q^{d\chi}$. If we substitute $1/(q^d T)$ into each factor of the cohomological expression for $Z(X_0, T)$, duality turns the factor in degree i into the factor in degree $2d - i$ times an explicit scalar.

We apply Theorem 10.21* with $K = \mathbf{Q}_\ell$ and X proper. It gives $b_{2d-i} = b_i$, and it shows that the eigenvalues of F_{2d-i} are the q^d/α for the eigenvalues α of F_i, with multiplicities. Hence $\det F_i \cdot \det F_{2d-i} = q^{db_i}$. Since $(-1)^i = (-1)^{2d-i}$, we get

$$D^2 = \prod_i (\det F_i \cdot \det F_{2d-i})^{(-1)^i} = q^{d\chi}.$$

Since $b_{2d-i} = b_i$, the terms of χ for i and $2d - i$ are equal and cancel modulo 2, so $\chi \equiv b_d \pmod 2$. If d is even, $d\chi$ is even anyway. If d is odd, the pairing on H^d of each connected component of X is perfect and alternating (Corollary 10.17*), so b_d is even. In both cases $d\chi$ is even. Then $\varepsilon^2 = 1$ because $D^2 = q^{d\chi}$.

Now consider a single factor. Since F_i is invertible, its eigenvalues α are nonzero. Pulling $-\alpha/(q^d T)$ out of each linear factor gives

$$\det(1 - q^{-d} T^{-1} F_i) = \prod_\alpha \Big(-\frac{\alpha}{q^d T}\Big)\Big(1 - \frac{q^d}{\alpha} T\Big) = (-1)^{b_i} \frac{\det F_i}{(q^d T)^{b_i}} \det(1 - T F_{2d-i}).$$

By Corollary 12.31, and since $H^i_c = H^i$ as X_0 is proper, $Z(X_0, T)$ is the product of the $\det(1 - TF_i)^{(-1)^{i+1}}$. We substitute $1/(q^d T)$ for T and apply the last display to each factor. Since

$(-1)^i = (-1)^{2d-i}$, the factors $\det(1 - TF_{2d-i})^{(-1)^{i+1}}$ multiply to $Z(X_0, T)$, and the scalar factors multiply to

$$(-1)^\chi D^{-1}(q^d T)^\chi = (-1)^\chi D\, T^\chi,$$

by $D^2 = q^{d\chi}$. The second form of the equation is the definition of ε. ∎

For the elliptic curve of Example 12.32 we have $d = 1$ and $b_0 = b_2 = 1$, $b_1 = 2$. So $\chi = 0$ and $D = \det(F_1)^{-1}\det(F_2) = 3/3 = 1$. The corollary says $Z(C_0, 1/(3T)) = Z(C_0, T)$, and indeed

$$\frac{1 + 3(3T)^{-2}}{(1 - (3T)^{-1})(1 - T^{-1})} = \frac{(3T^2 + 1)/(3T^2)}{(3T - 1)(T - 1)/(3T^2)} = \frac{1 + 3T^2}{(1 - 3T)(1 - T)}.$$

Example 12.34 (The functional equation of the projective line)**.** For $X_0 = \mathbf{P}^1_{\mathbf{F}_q}$ we have $d = 1$, $b_0 = b_2 = 1$, $b_1 = 0$, so $\chi = 2$, and $D = \det(F_0)\det(F_2) = 1 \cdot q = q$, so that $\varepsilon = 1$. The corollary predicts $Z(\mathbf{P}^1, 1/(qT)) = qT^2 Z(\mathbf{P}^1, T)$. Directly,

$$\frac{1}{(1 - (qT)^{-1})(1 - T^{-1})} = \frac{qT^2}{(qT - 1)(T - 1)} = \frac{qT^2}{(1 - T)(1 - qT)}.$$

Here the factor T^χ is not trivial, unlike for the elliptic curve, and the constant $q^{d\chi/2} = q$ comes from the twist (-1) on H^2.

Remark 12.35 (Duality and weights)**.** The functional equation is an identity of rational functions, not of power series at $T = 0$, since $1/(q^d T)$ is not a power series in T.

What duality does not give is the absolute value of the eigenvalues α. In Chapters 13*–Chapter 19* we prove that the eigenvalues of F_i have absolute value $q^{i/2}$ under every embedding of $\overline{\mathbf{Q}}_\ell$ into $\mathbf{C}$. It follows that the factors $\det(1 - TF_i)$ for different i have no common zeros, and hence we can read them off from $Z(X_0, T)$ alone. In Chapter 19* we use this to show that they are polynomials with integer coefficients that do not depend on ℓ.

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